WiL Update Progress

Vision-Based Navigation Pipeline for AMRs in Dynamic Warehouses

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1. Problem Statement

- **Identified Gensurv's issues**

Current Solutions

LMR : Latent Mobile Robot

↓

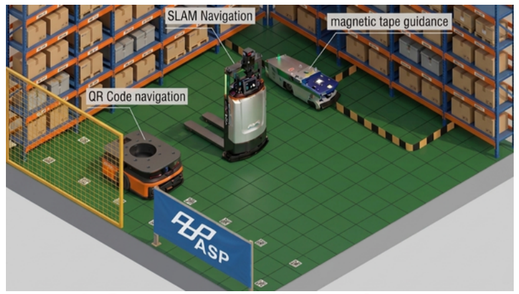
QR-navigated robot

PLY AGV Autonomous Tow truck

↓

2D LiDAR Navigation

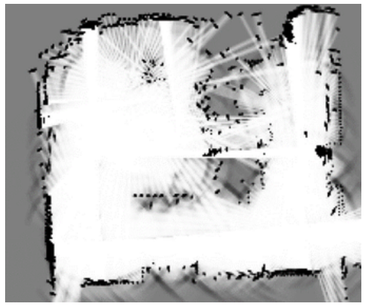
Limitations



Infrastructure dependence

Problem

High cost on changing warehouse infrastructure



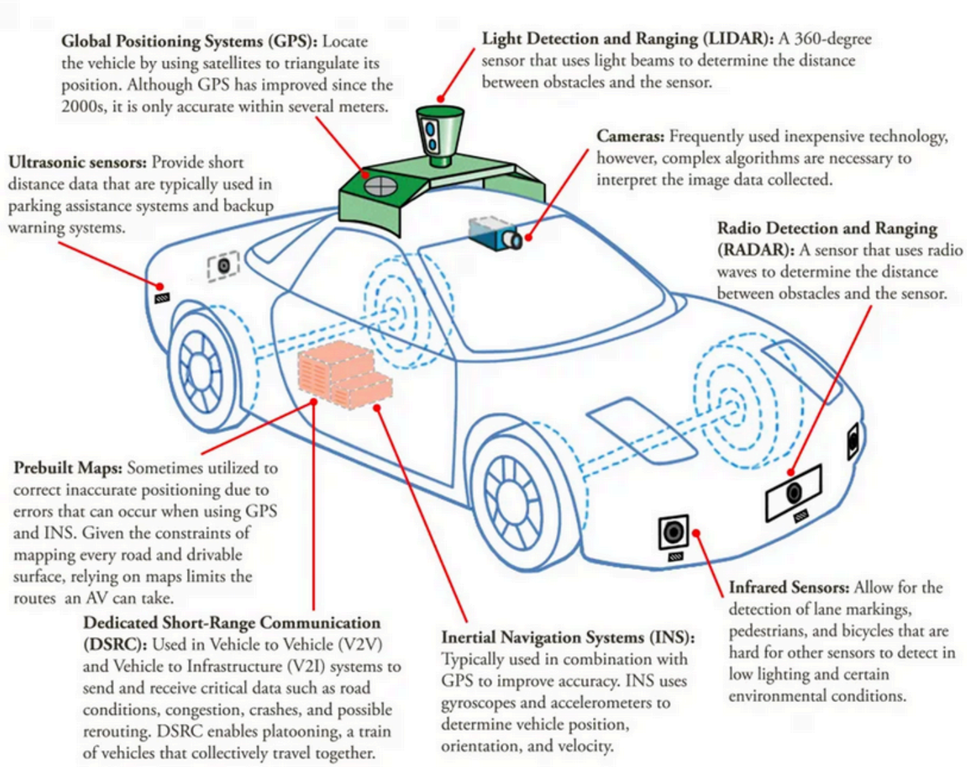
Low Robustness for different environment

https://gensurv.com/products-th/

1. Problem Statement

- **Why do we use camera-based SLAM ?**

Examples of Technologies Supporting Autonomous Vehicles



Source: Center for Sustainable Systems, University of Michigan (2024)

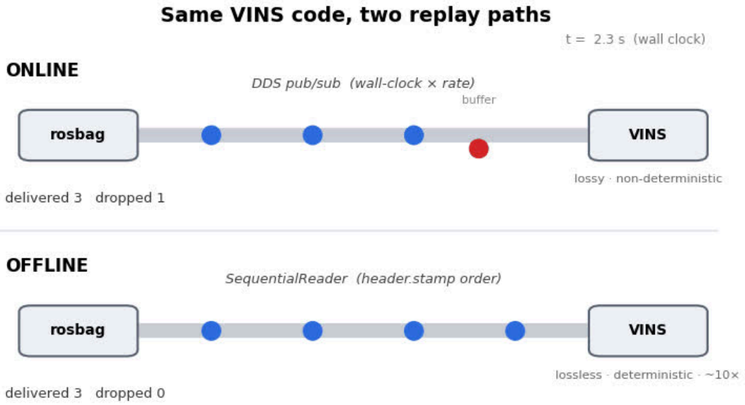
Modality	Affected by Illumination	Affected by weather	Color	Depth	Range	Accuracy	Size	Cost
Lidar	-	✓	-	✓	medium (< 200m)	high	large*	high*
Radar	-	-	-	✓	high	medium	small	medium
Ultrasonic	-	-	-	✓	short	low	small	low
Camera	✓	✓	✓	-	-	-	smallest	lowest
Stereo Camera	✓	✓	✓	✓	medium (< 100m)	low	medium	low
Flash Camera [77]	✓	✓	✓	✓	medium (< 100m)	low	medium	low
Event Camera [78]	limited	✓	-	-	-	-	smallest	low
Thermal Camera [79], [80]	-	✓	-	-	-	-	smallest	low

Sensor	Estimated Cost per unit (USD)
Lidar	500 - 75000
Camera	20 - 500 (required 8-12 cameras)

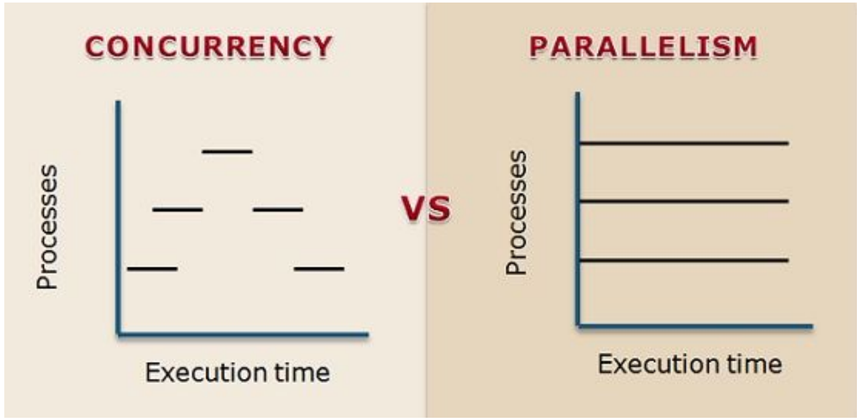
https://www.krungsri.com/en/research/research-intelligence/ai-autonomous-car-2024,
https://patentpc.com/blog/the-cost-of-self-driving-technology-how-much-do-av-components-really-cost-market-breakdown

1. Problem Statement

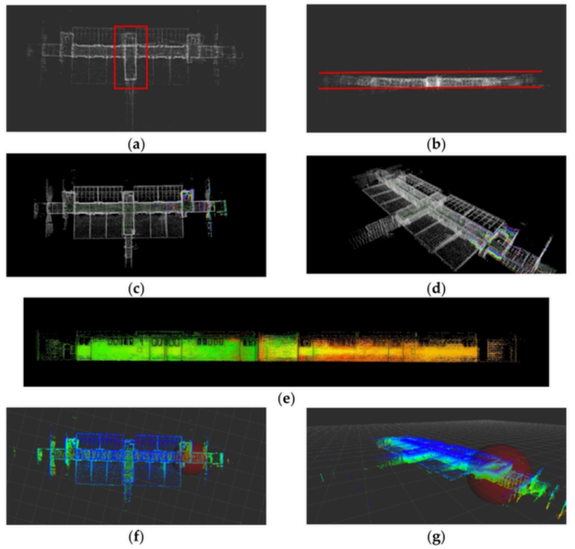
- **Implementation challenges**



Timeliness Challenge



The Concurrency Challenge

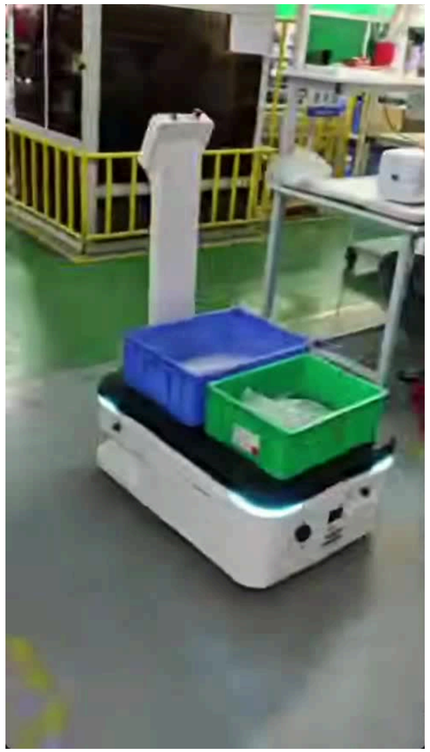


Context awareness Challenge

SOFIYA SEMENOVA; STEVEN KO; YU DAVID LIU; LUKASZ ZIAREK; KARTHIK DANTU; A Comprehensive Study of Systems Challenges in Visual Simultaneous Localization and Mapping Systems; 2024

1. Problem Statement

While dynamic obstacles in the warehouse are indeed present, the warehouse layout, material flow patterns, and autonomous robot behaviors remain unknown.



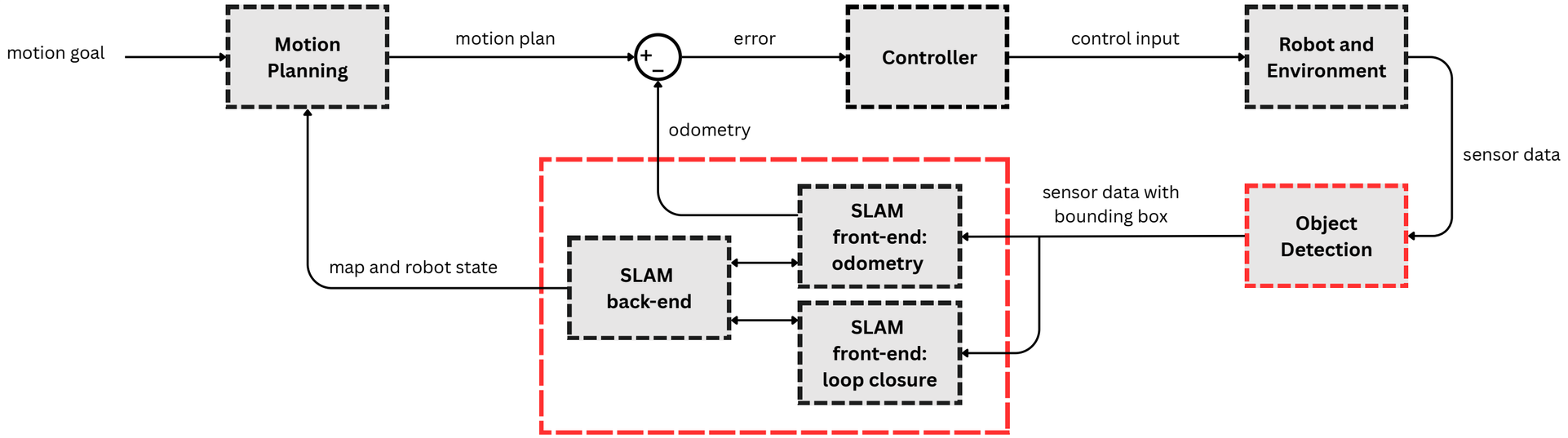
Dynamic Obstacles

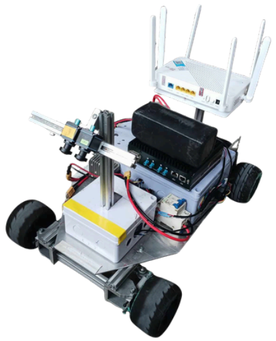
Map Restructure

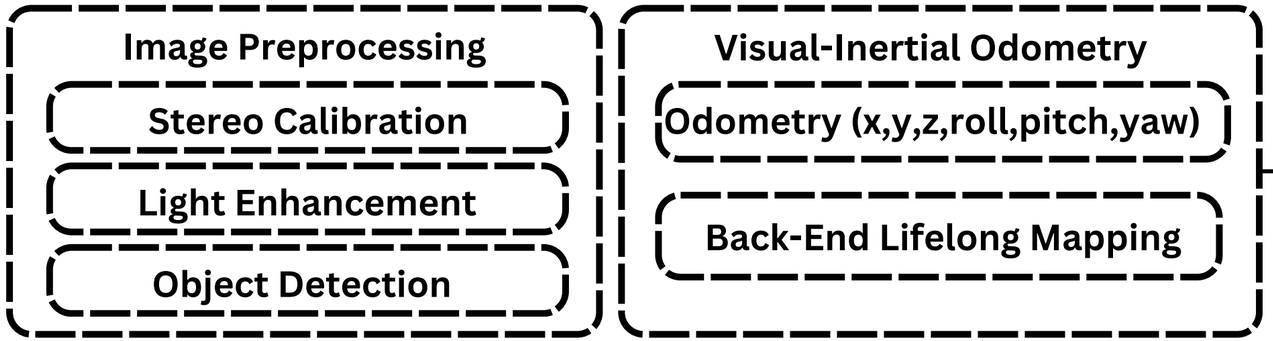
- https://www.google.com/url?sa=t&source=web&rct=j&url=https%3A%2F%2Fwww.buske.com%2Fblog%2Ffood-grade-warehousing-a-complete-guide-to-selecting-a-provider&ved=0CBkQjhxqGAoTCJiZ0Lm--pYDFQAAAAAdAAAAABCrAQ&opi=89978449
- https://www.youtube.com/watch?v=KJR6aj8f5VA
- https://ssrudlp.ssru.ac.th/data-file/teacher_work/file/c3d596723eef77c690f777bff46f0439.pdf

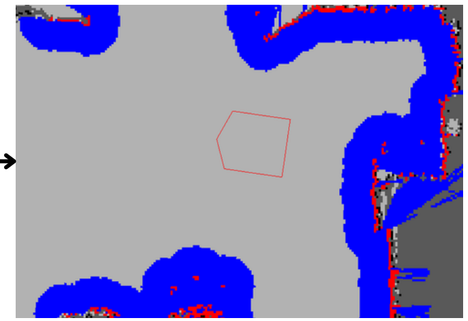
2. Requirement and Scope

Requirement









2D costmap

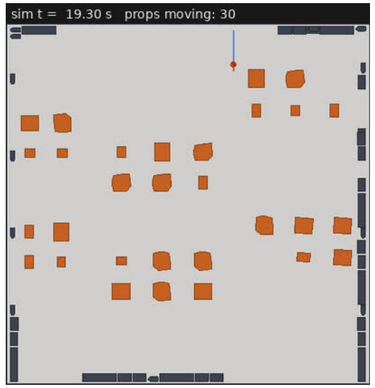
ATE <= 0.05 m

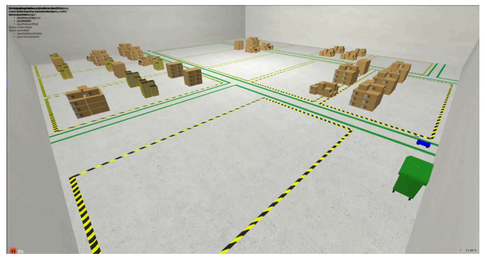
RPE (delta t = 1 s) <= 1.5 m

2. Requirement and Scope

Scope

Simulation

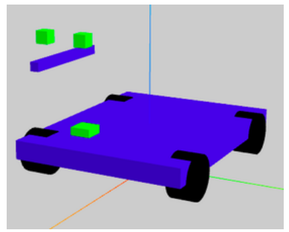




GAZEBO

Dynamic object

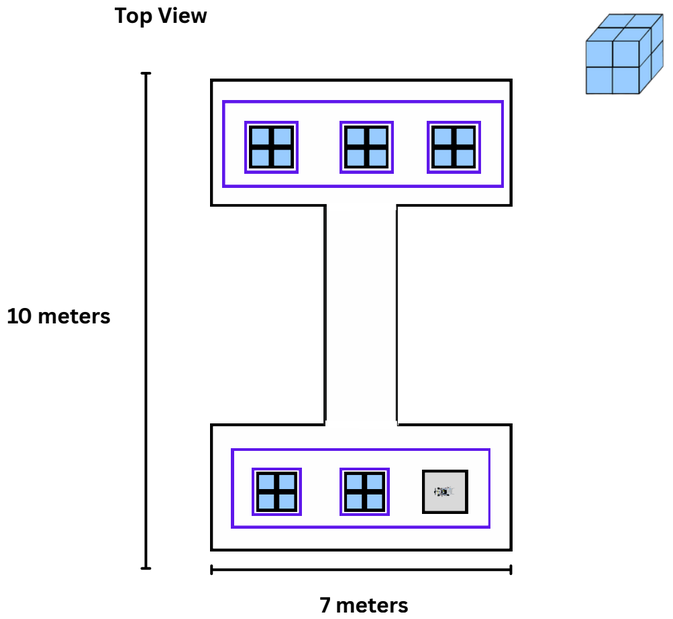
ROS2



Maximum robot speed = 1.5 m

Real world

Gensurv building 1st floor



Hardware

3. Literature Review

Problem : Memory & Computational Explosion

- Future direction of visual SLAM
- Visual SLAM comparison

Solution :

Visual Odometry

- VINS-fusion
- ORB-SLAM3

Mapping Back-End

- RTAB-MAP

Problem : Dynamics environment

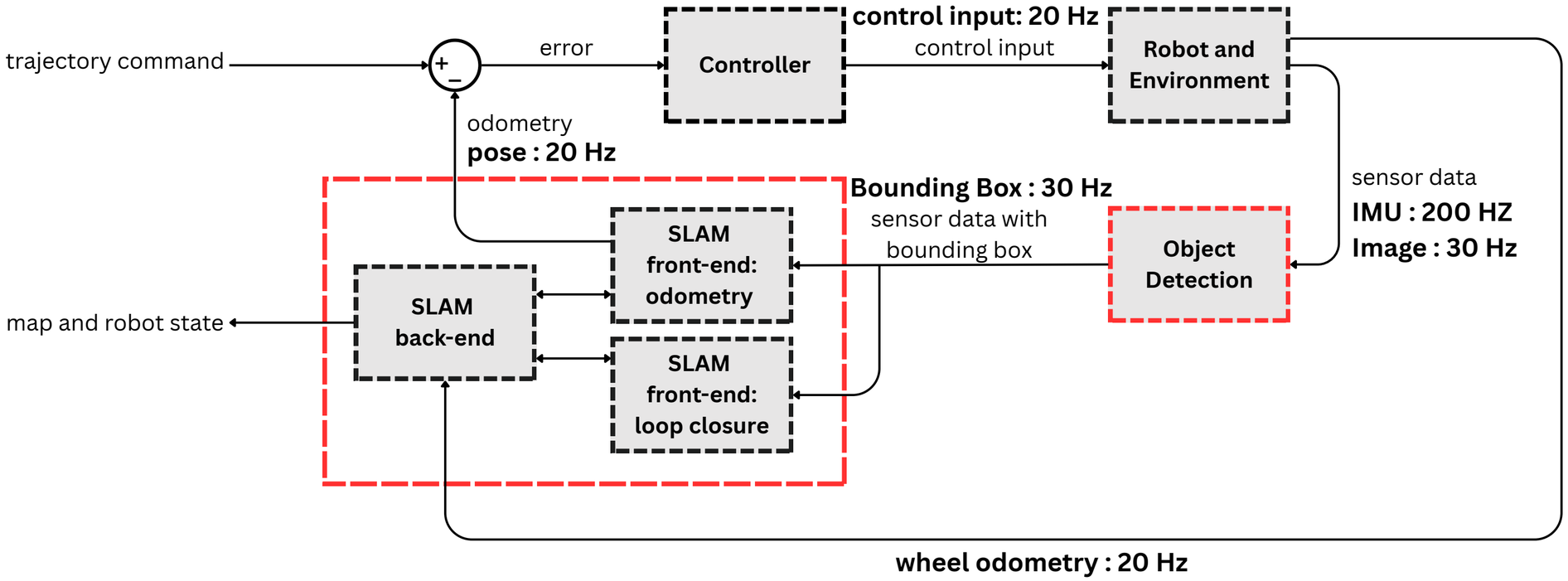
- Survey paper of Structure from Motion

Solution : Semantic Understanding

- Survey paper of semantic visual SLAM
- Survey paper of object detection model
- RY-SLAM

4. Methodology

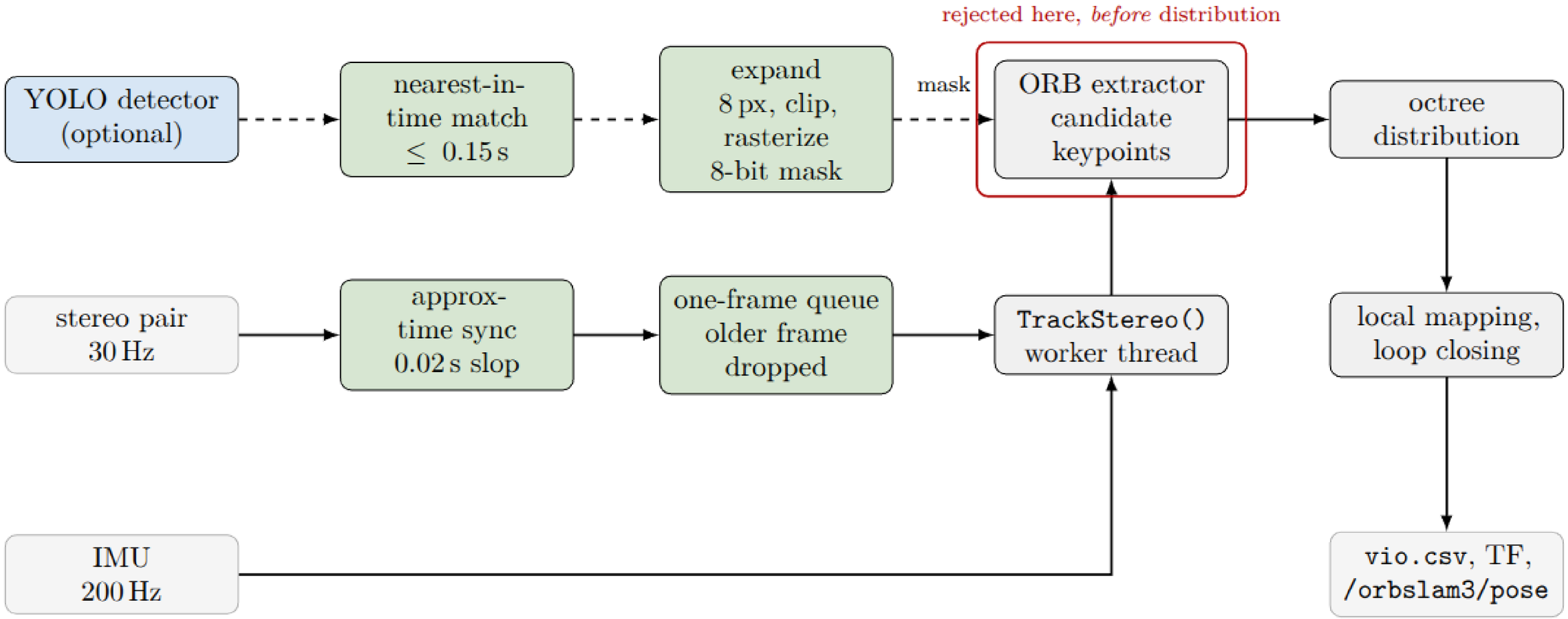
- **System Architecture**



4. Methodology

- **System Architecture : ORBSLAM3 Approach**

Object Detection Integration



4. Methodology

- **System Architecture : ORBSLAM3 Approach**

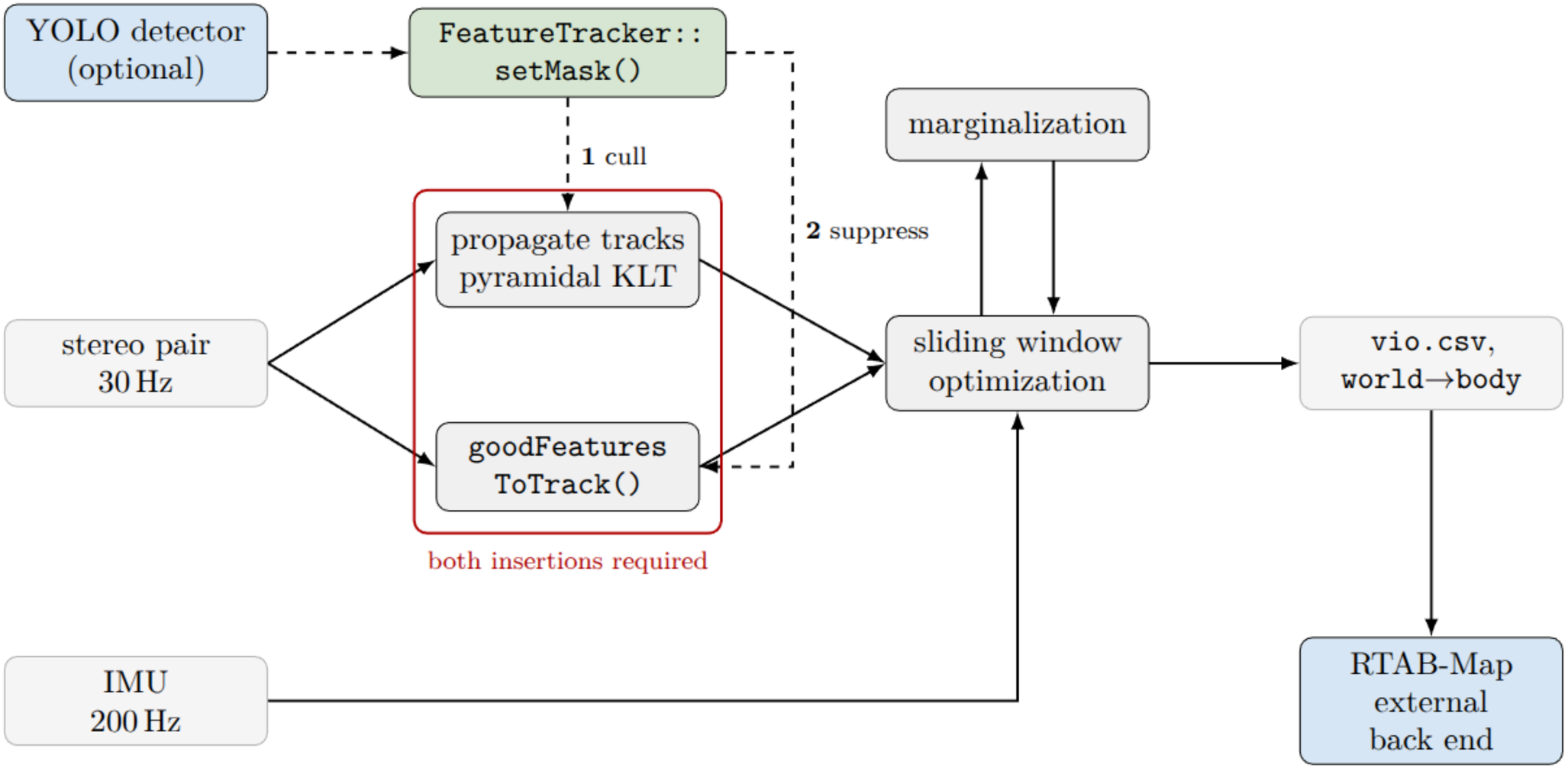
Object Detection Integration

Parameter	Value	Role
Wrapper (built here)		
Stereo sync slop	0.02 s	approximate-time pairing
Image queue depth	1 frame	older frame dropped, drop count logged
Detector match window	0.15 s	nearest in source-image time
Box expansion	8 px	before rasterizing the mask
Mask reject threshold	0.8 of frame	safety cut-out on runaway coverage
Boundary rule	≥ 5 of 3 × 3	candidate rejected when dynamic
Masked camera	left only	stereo matches form from left keypoints
Extractor and core		
ORBextractor.nFeatures	1500	per-frame feature budget
ORBextractor.nLevels	8	pyramid levels
ORBextractor.scaleFactor	1.2	inter-level scale
ORBextractor.iniThFAST	20	initial corner threshold
ORBextractor.minThFAST	7	fallback threshold
Stereo.ThDepth	90	close/far stereo split
IMU.Frequency	200 Hz	preintegration rate
loopClosing	1	comment in the same file says "set to 0 deliberately"

4. Methodology

- **System Architecture : VINS + RTAB Approach**

Object Detection Integration



4. Methodology

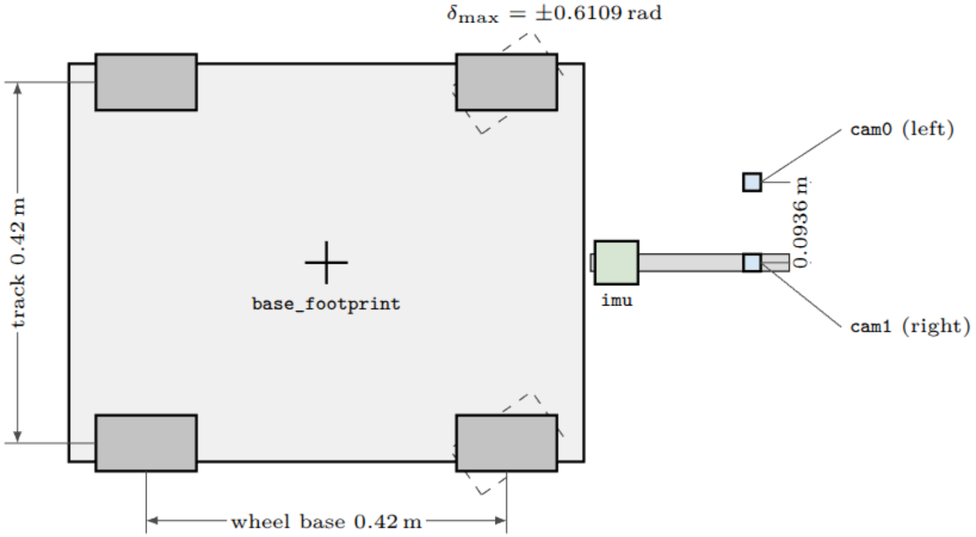
- **System Architecture : VINS + RTAB Approach**

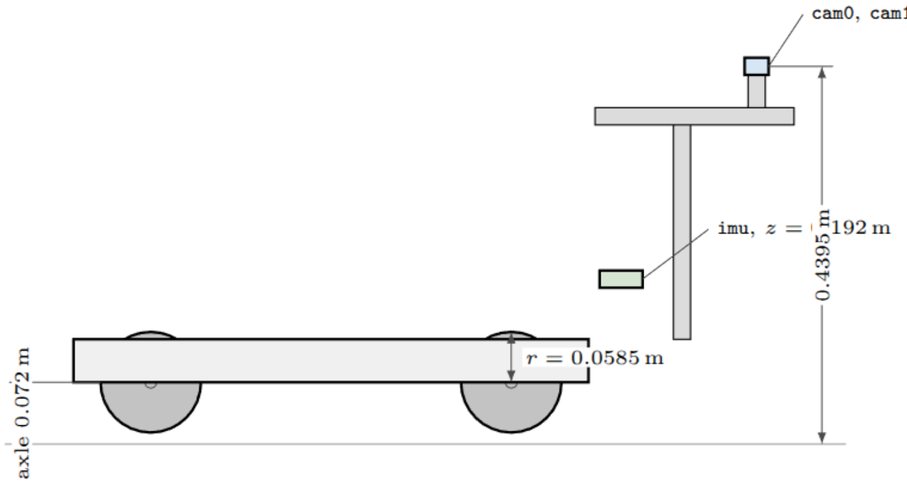
Object Detection Integration

Parameter	Value	Role
image_skip	1	every image processed
max_cnt	150	tracked-feature budget
min_dist	30 px	minimum feature separation
freq	30 Hz	target feature publication
keyframe_parallax	10 px	keyframe decision
F_threshold	1	RANSAC fundamental-matrix gate
max_solver_time	0.08 s	Ceres budget per update
max_num_iterations	100	Ceres iteration cap
estimate_extrinsic	0	simulated extrinsics held fixed
estimate_td	0	camera–IMU delay held at 0 s
g_norm	9.8 m/s.s	matches the world's gravity
Image queue	100	raised to survive initialization backlog
IMU queue	2000	raised for the same reason

5. Experimental Design

- **Simulation : Sensor**





- **Camera :** $depth = \frac{focal \times baseline}{disparity}$; $disparity = uL - uR$

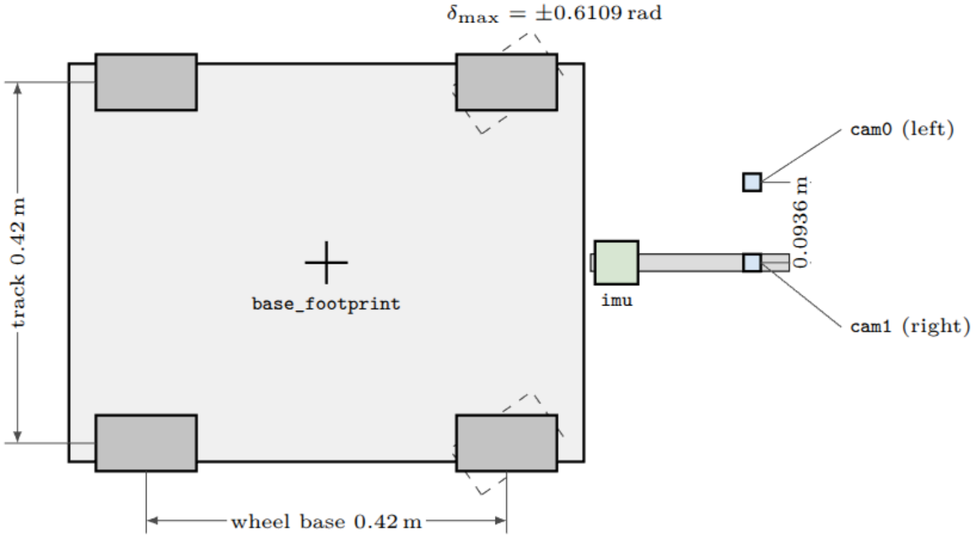
Parameter	cam0 (left)	cam1 (right)	imu
Resolution	1280 × 720	1280 × 720	-
Rate	30 Hz	30 Hz	200 Hz
Horizontal FOV	π/2 rad (90°)	π/2 rad (90°)	-
$f_x = f_y$	640	640	-
$(c_x = c_y)$	(640, 360)	(640, 360)	-
Distortion	zero	zero	-
Position from base_footprint (m)	(0.496, 0.0936, 0.4395)	(0.496, 0, 0.4395)	(0.338, 0, 0.192)
Position from IMU body (m)	(0.158, 0.0936, 0.2475)	(0.158, 0, 0.2475)	(0, 0, 0)
Stereo baseline	0.0936 m		-

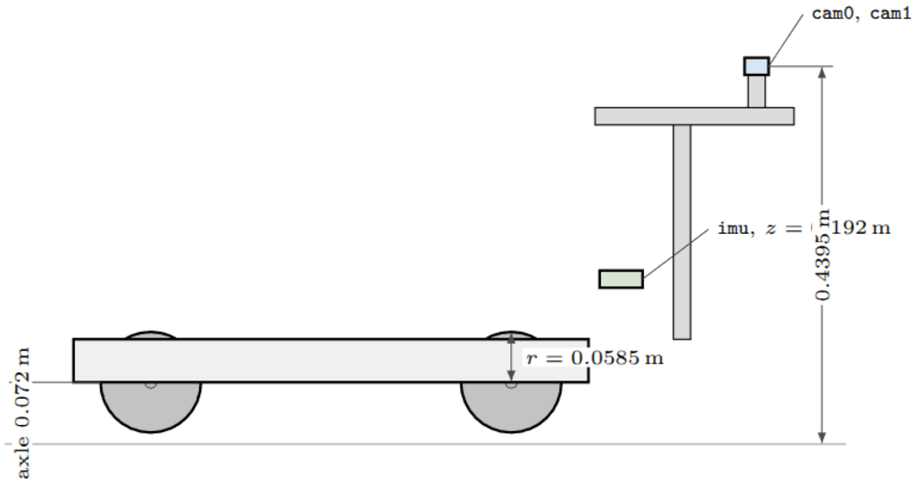
- **IMU :**

Parameter	Configured (VINS, ORB)	Measured from bag	Units
acc_n	0.02	0.0139–0.0294	$\frac{ms^{-2}}{\sqrt{Hz}}$
gyr_n	0.002	0.000300–0.000364	$\frac{rads^{-1}}{\sqrt{Hz}}$
acc_w	0.001	not measurable	$\frac{ms^{-3}}{\sqrt{Hz}}$
gyr_w	0.0001	not measurable	$\frac{rads^{-2}}{\sqrt{Hz}}$

5. Experimental Design

- **Simulation : Platform Geometry**

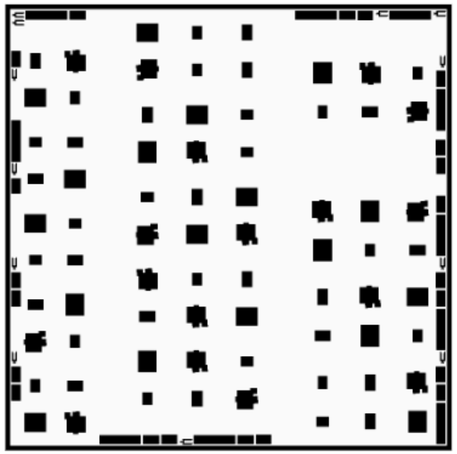




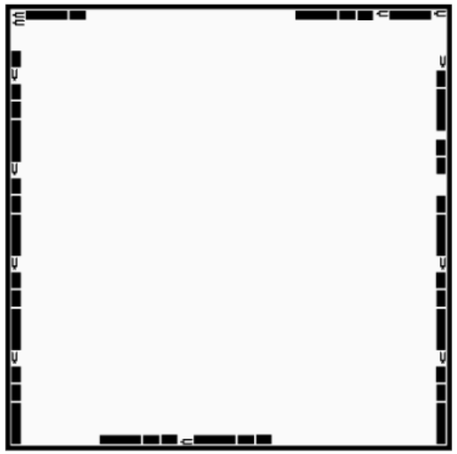
Quantity	Value
Wheel base (front to rear axle)	0.42 m
Track (wheel separation)	0.42 m
Kingpin width	0.42 m
Wheel radius	0.0585 m
Wheel width	0.065 m
Steering limit	Approx. 0.6109 rad
Chassis mass	10.1 kg
Maximum speed	1.5 m/s
Maximum acceleration	0.45 m/s²
Odometry publish rate	50 Hz
Joint-state publish rate	1 kHz

5. Experimental Design

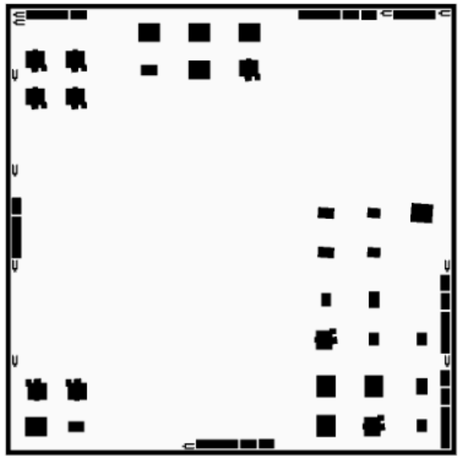
Obstacles



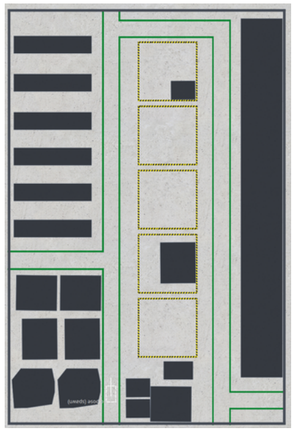
Static



Empty floor



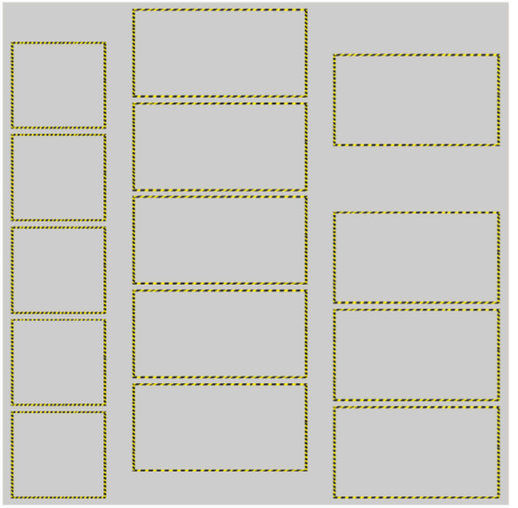
Dynamic



Tiny map

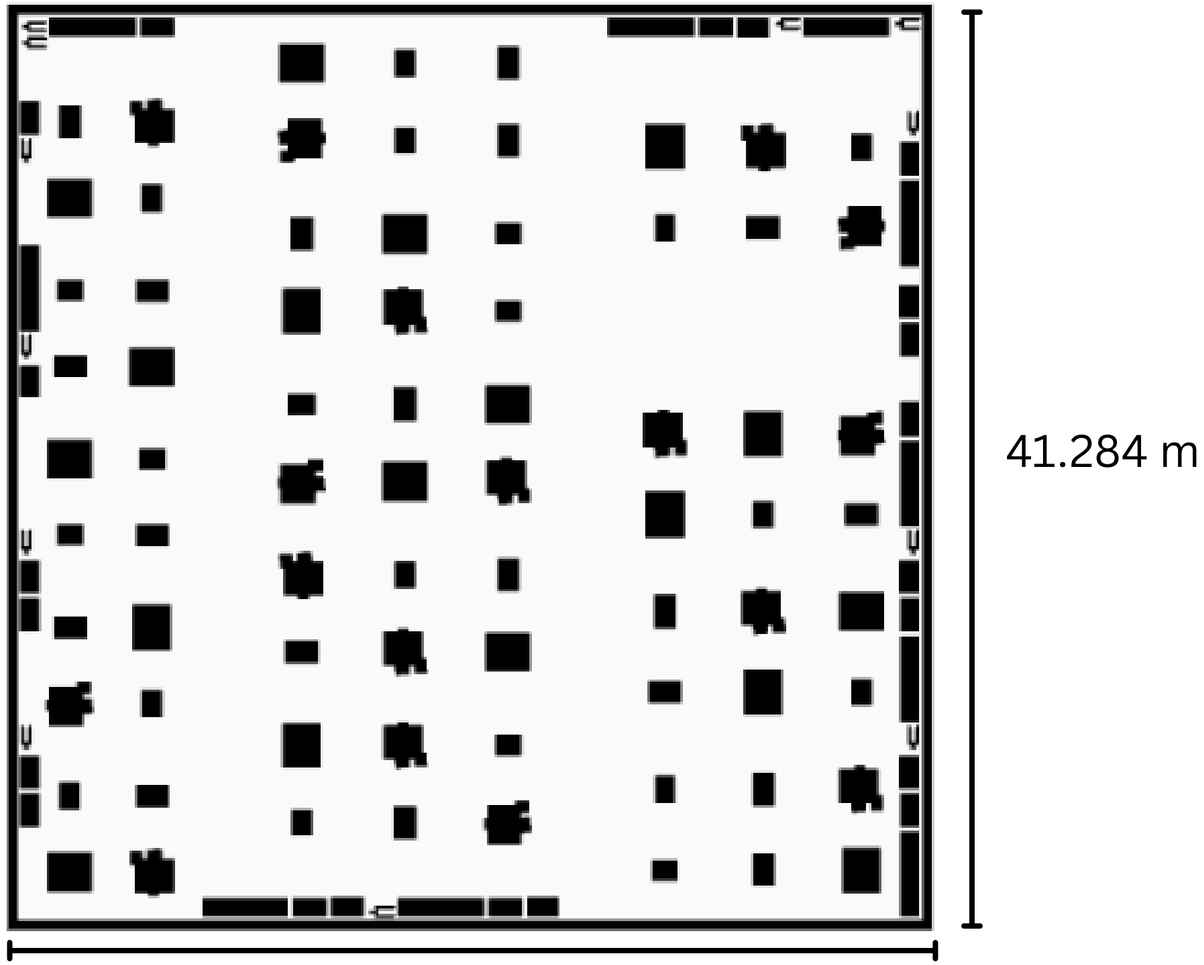
Floor texture

No texture

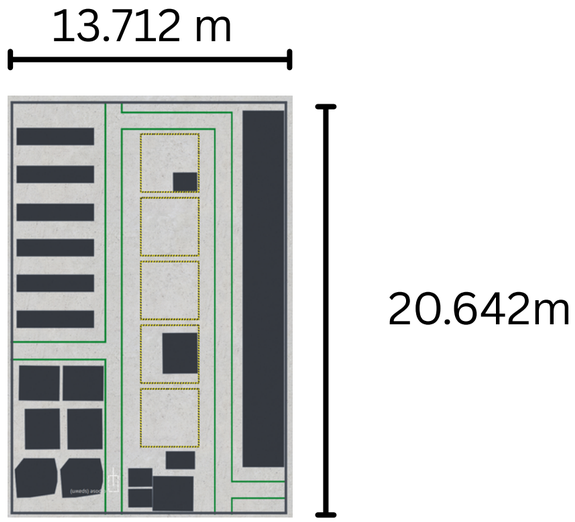


Bay zone

5. Experimental Design

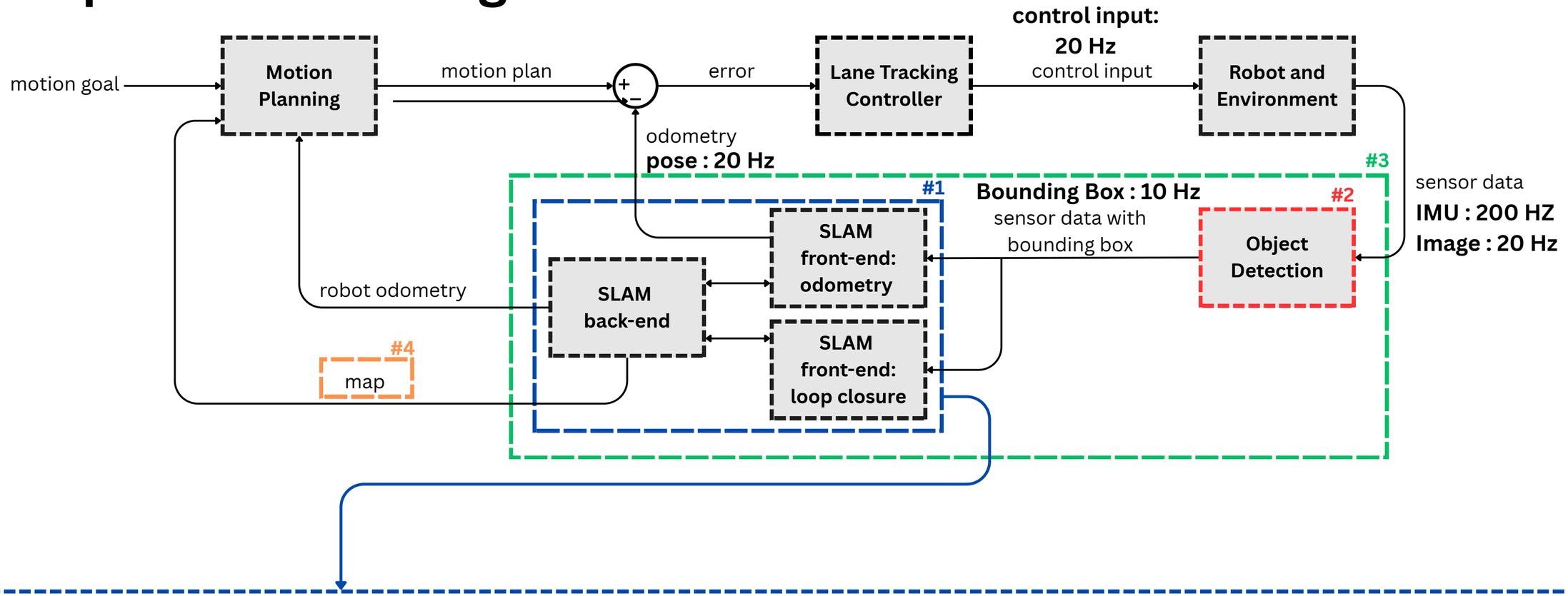


Large map



Tiny map

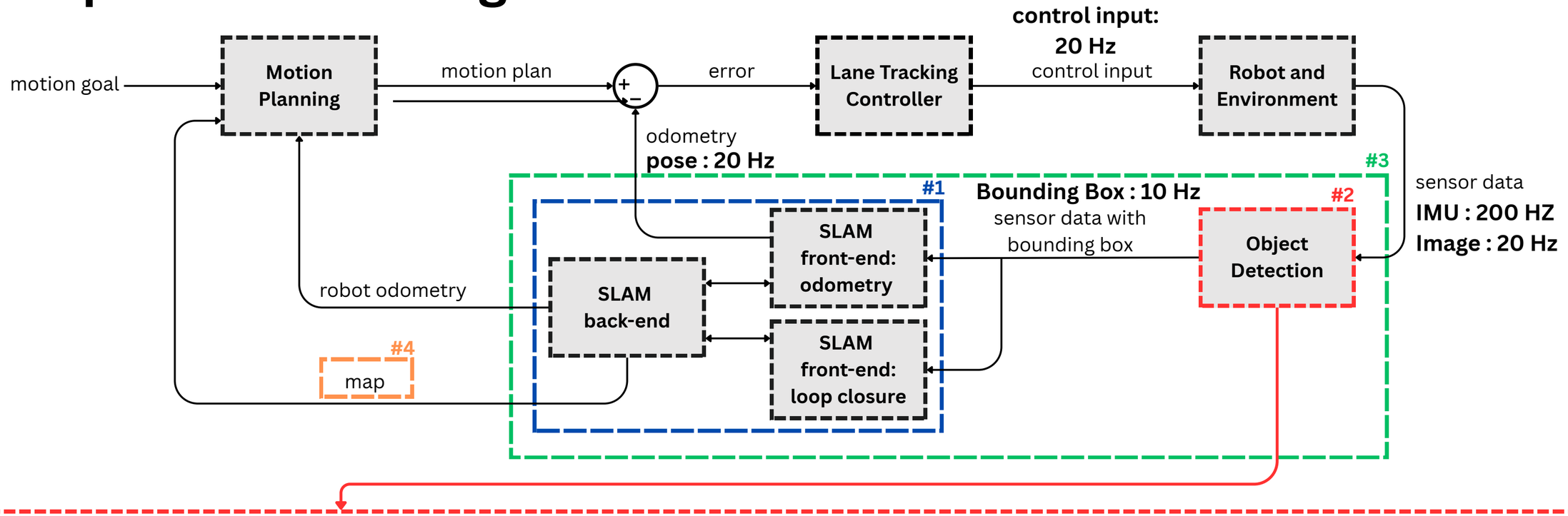
5. Experimental Design



<u>Experiment 1 : Non-Visual Baseline and Estimator Comparison</u>

- **Objective :** Establish a dead-reckoning baseline to determine if visual pipelines (ORB-SLAM3 and VINS-Fusion) genuinely outperform non-visual ones (wheel odometry, IMU, and wheel+IMU EKF).

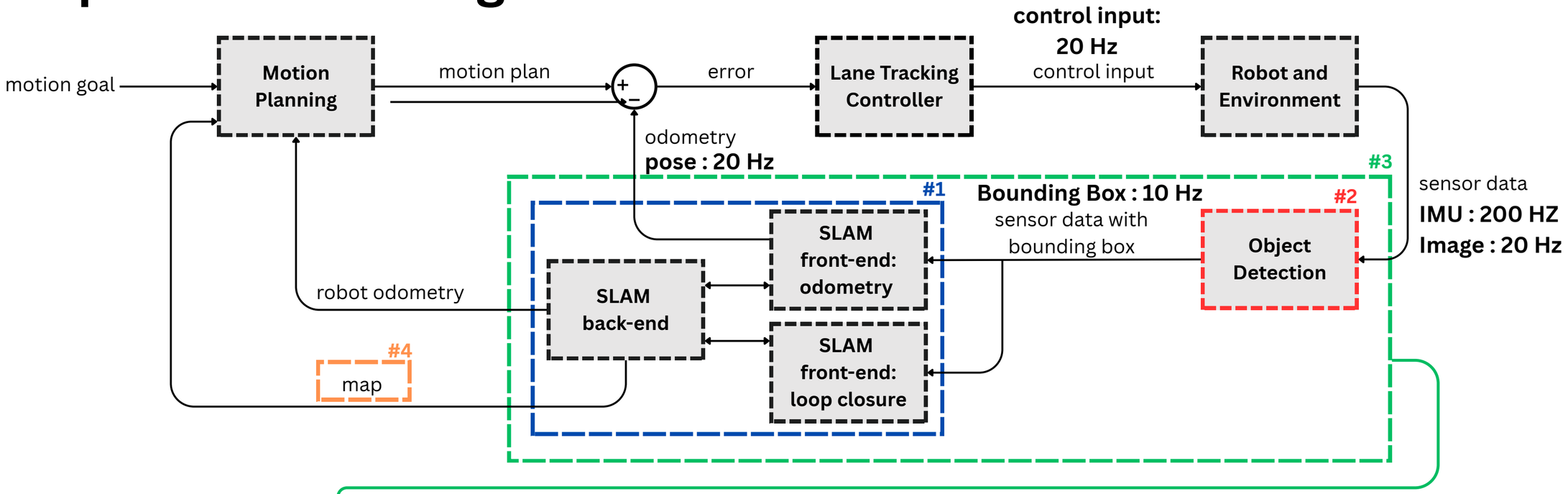
5. Experimental Design



Experiment 2 : Offline YOLO Detector Performance

- **Objective :** Measure standalone detector performance independent of SLAM so future filtering results are attributed to mask logic rather than detector capabilities.
- **Analysis**: Assess how reliably the detector localizes and names warehouse objects.
- **Constraints**: Measure image-area coverage to determine the maximum amount of the frame that semantic filtering could theoretically remove.

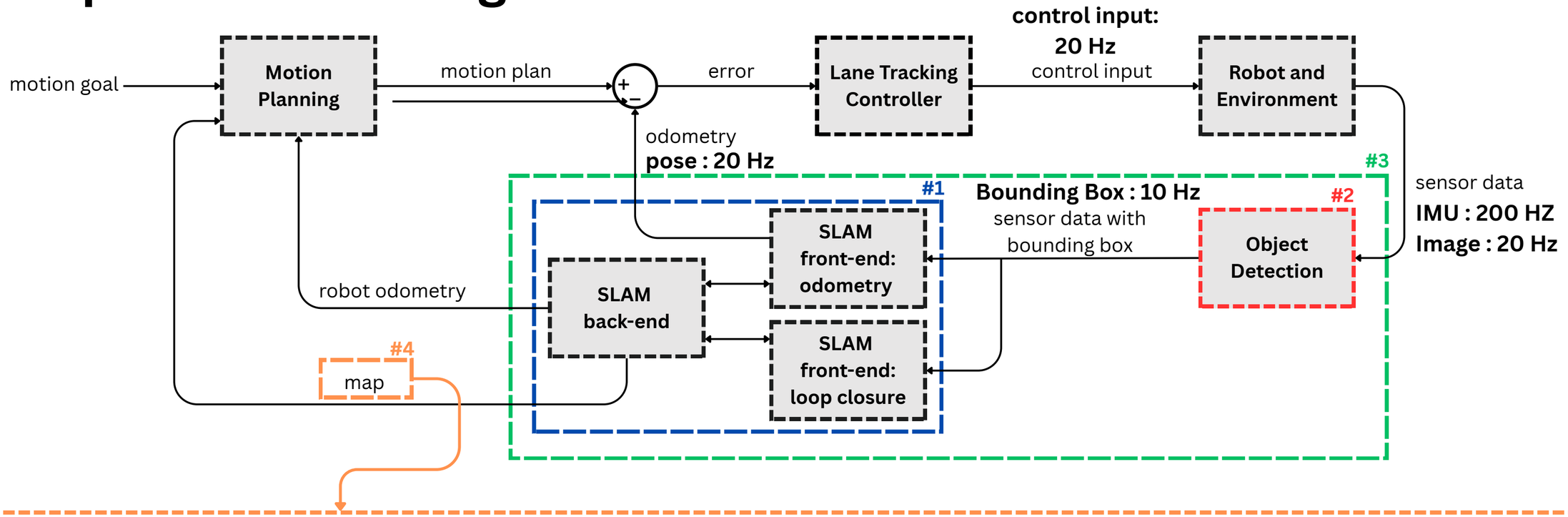
5. Experimental Design



Experiment 3 : Visual-Inertial Odometry With Dynamic Environment Filtering Evaluation

- **Objective :** Determine if masking dynamic-object features improves localization and measure the associated computational costs.
- **Constraints**: Verify that tracking survives the filtering and that added detector costs do not push pipelines below the input frame rate.

5. Experimental Design



Experiment 4 :Dense Map Reconstruction and Map-Quality Evaluation

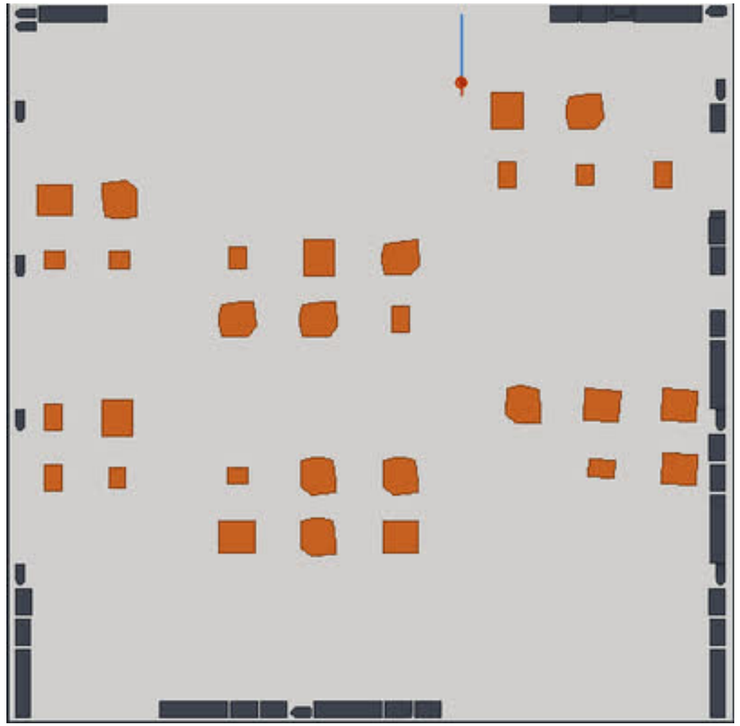
- **Objective :** Verify if the reconstructed map accurately reflects the physical environment and is usable by a route planner.
- **Constraints**: Score drivability against a known, successfully driven route so any logged collisions represent proven mapping errors rather than assumptions.

5. Experimental Design

#1 Experiment 1 : Non-Visual Baseline and Estimator Comparison

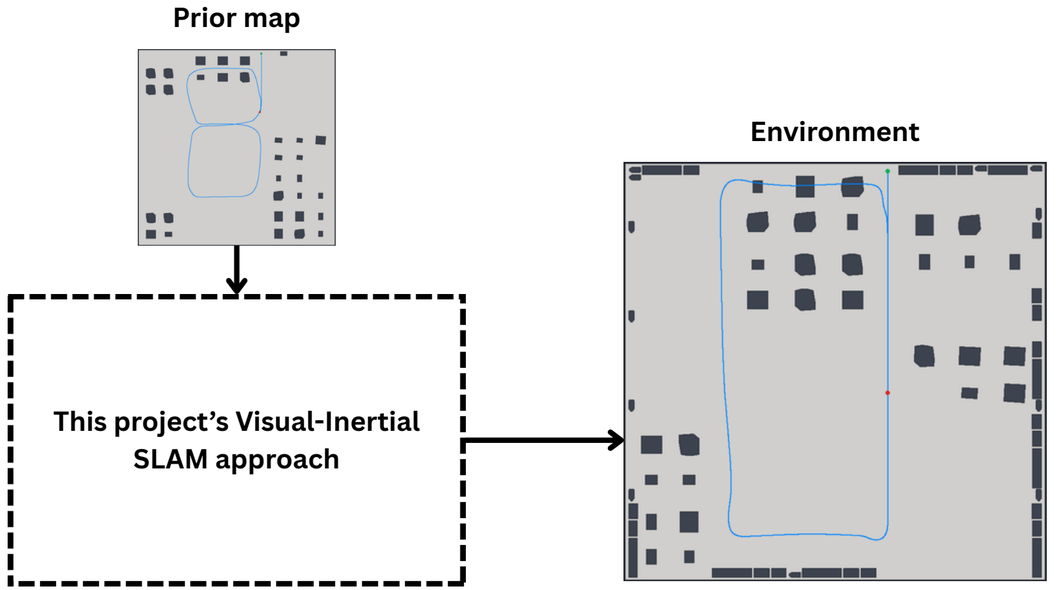
#2 Experiment 2 : Offline YOLO Detector Performance

#3 Experiment 3 : Visual-Inertial Odometry With Dynamic Environment Filtering Evaluation



Dynamic Obstacles

#4 Experiment 4 :Dense Map Reconstruction and Map-Quality Evaluation

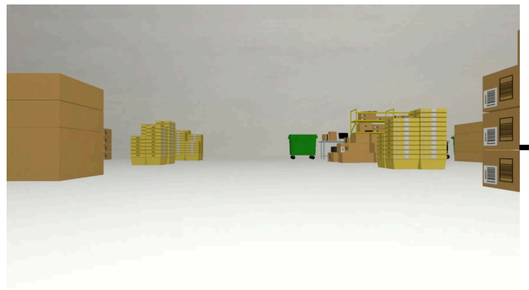


Map Reconstruction

5. Experimental Design

Experiment 1 : Non-Visual Baseline and Estimator Comparison

- Process :



1.Collect datasets

2. Measure noise using gt odometry

wheels only
IMU only
wheels + IMU fused by a filter
VINS-Fusion (cameras + IMU)
ORB-SLAM3 (cameras + IMU)

3. Reconstruct the route

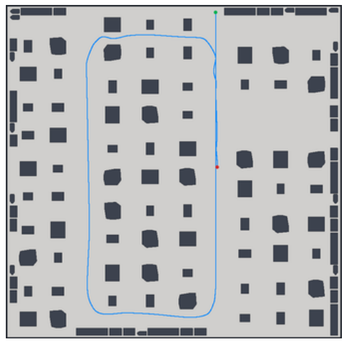
- Evaluation Metric:
  - Aligned ATE RMSE
  - anchored RMS and final error

5. Experimental Design

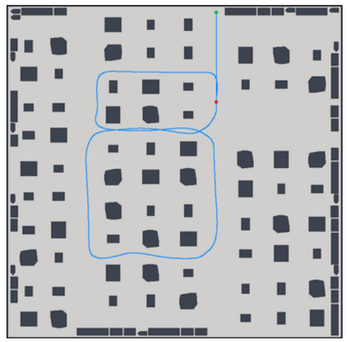
Experiment 1 : Non-Visual Baseline and Estimator Comparison

- Dataset information

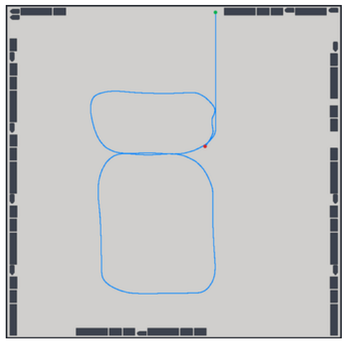
Dataset	Family	Size (MB)	Duration. (s)	Messages	/clock	Truth	Prop.
dataset_allsensor_000	static_01	689	129.3	211 811	✓	✓	✓
dataset_allsensor_001	static_01	674	126.2	198 873	✓	✓	✓
dataset_allsensor_002	static, wall only	750	105.5	215 528	✓	✓	✓
dataset_allsensor_003	bayline, wall only	1056	141.2	255 699	✓	✓	✓
dataset_allsensor_004	bayline, distributed	1311	168.6	253 942	✓	✓	✓



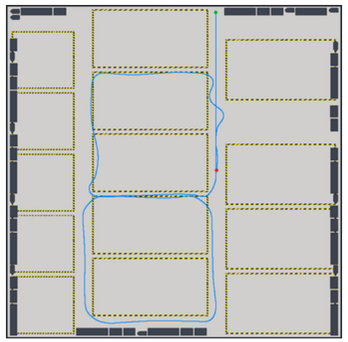
dataset_allsensor_000



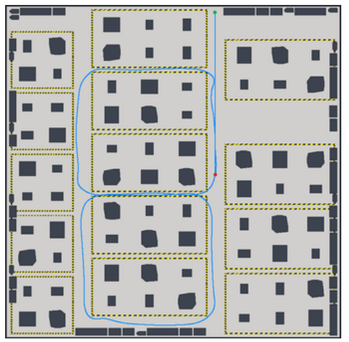
dataset_allsensor_001



dataset_allsensor_002



dataset_allsensor_003



dataset_allsensor_004

4. Methodology

- **System Architecture : EKF**

Estimator	Inputs	State	Correction
Wheel odometry	rear wheel speeds, steering angles (1 kHz)	$[p_x, p_y, \theta]$	none
IMU dead reckoning	gyroscope z, accelerometer x (200 Hz)	$[p_x, p_y, \theta, v]$	none
EKF wheel + IMU	both of the above	$[p_x, p_y, \theta, v, b_w]$	wheel speed corrects v; $(\omega_{gyro} - \omega_{wheel})$ corrects b_ω

- Prediction state

$$p_x \mathrel{+}= v\cos\theta_m\,\Delta t + \tfrac{1}{2}a\cos\theta_m\,\Delta t^2,$$
$$p_y \mathrel{+}= v\sin\theta_m\,\Delta t + \tfrac{1}{2}a\sin\theta_m\,\Delta t^2,$$
$$v \mathrel{+}= a\,\Delta t, \qquad \theta \mathrel{+}= \omega\,\Delta t.$$

midpoint heading $\theta_m = \theta + \tfrac{1}{2}\omega\,\Delta t$

- Update state

$$\mathbf{z} = \begin{bmatrix} v_{\text{wheel}} \\ \omega_{\text{gyro}} - \omega_{\text{wheel}} \end{bmatrix}, \qquad \mathbf{H} = \begin{bmatrix} 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 1 \end{bmatrix}, \qquad \mathbf{R} = \operatorname{diag}\left(\sigma_{v,w}^2,\ \sigma_{\omega,w}^2 + \sigma_g^2\right)$$

$$\mathbf{P} \leftarrow (\mathbf{I} - \mathbf{KH})\mathbf{P}(\mathbf{I} - \mathbf{KH})^\top + \mathbf{KRK}^\top$$

5. Experimental Design

Experiment 1 : Non-Visual Baseline and Estimator Comparison

- Noise Result

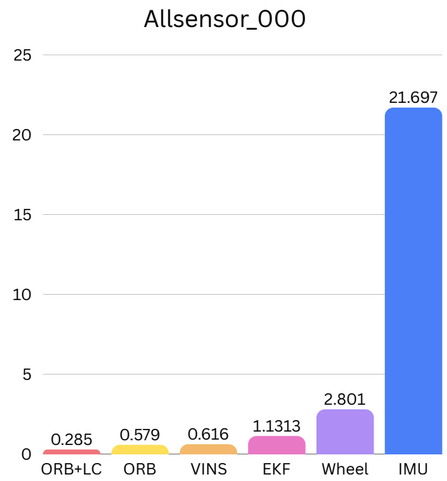
Group	Quantity	_000plain	_001plain	_002plain	_003bayline	_004bayline	Units
IMU	gyroscope z bias	0.0000648	0.0000775	0.0000928	0.000609	0.0000258	rad/s
	gyroscope σ	0.00436	0.00424	0.00481	0.00503	0.00515	rad/s
	gyroscope z scale	0.999	1	1.0002	1.0002	1	–
	accelerometer x bias	0.0191	0.0304	0.0683	0.0309	0.0016	m/s.s
	accelerometer x scale	1.0199	1.0249	1.0669	1.0525	1.0465	–
	measured g	9.7809	9.8191	9.7827	9.8037	9.8346	m/s.s
Wheel	speed scale	1.0002	0.9985	1.0005	1.0071	1.0001	–
	speed σ	0.0407	0.0484	0.0496	0.0607	0.0505	m/s
	yaw-rate scale, bicycle	1.011	0.9527	1.0034	0.9375	0.925	–
	yaw-rate σ bicycle	0.0659	0.0804	0.0705	0.1192	0.0613	rad/s
	yaw-rate scale, differential	1.2162	1.1442	1.2194	1.2748	1.1404	–
	yaw-rate σ differential	0.2359	0.3074	0.262	0.3477	0.2828	rad/s
Window	static duration	0.68	0.58	1.54	1.04	0.52	s
	maximum tilt	0.0016	0.0018	0.0015	0.0018	0.0017	degree

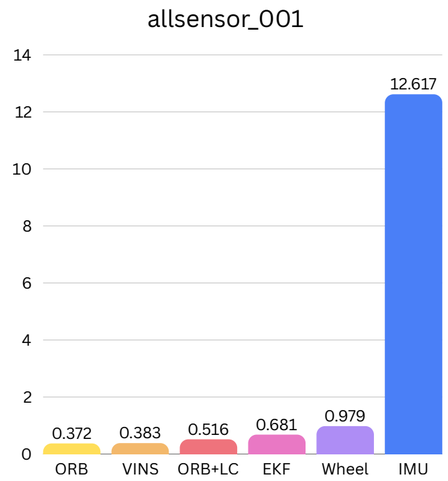
	Ours	EuRoC	
gyro noise density	0.000339	0.00017	**2× noisier**
accel noise density	0.004	0.002	**2× noisier**
IMU rate	200 Hz	200 Hz	**same**

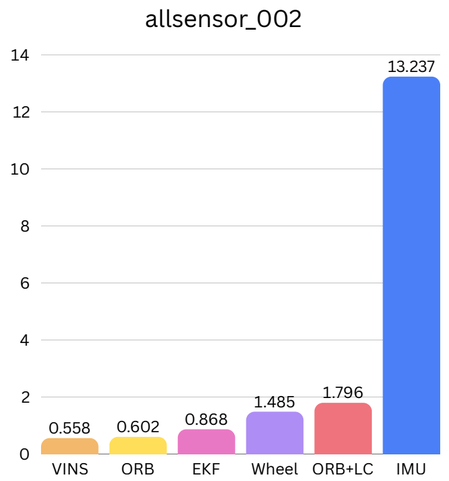
5. Experimental Design

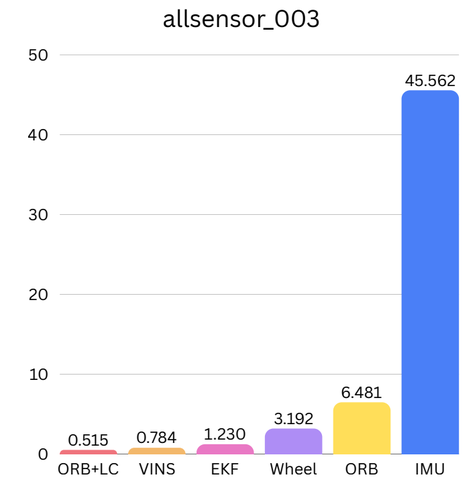
Experiment 1 : Non-Visual Baseline and Estimator Comparison

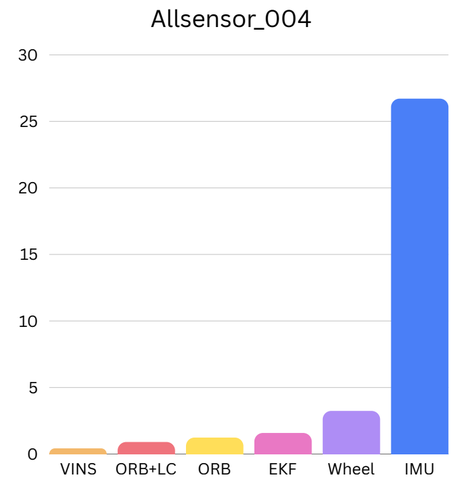
- Aligned Absolute Trajectory Error (ATE in meters)









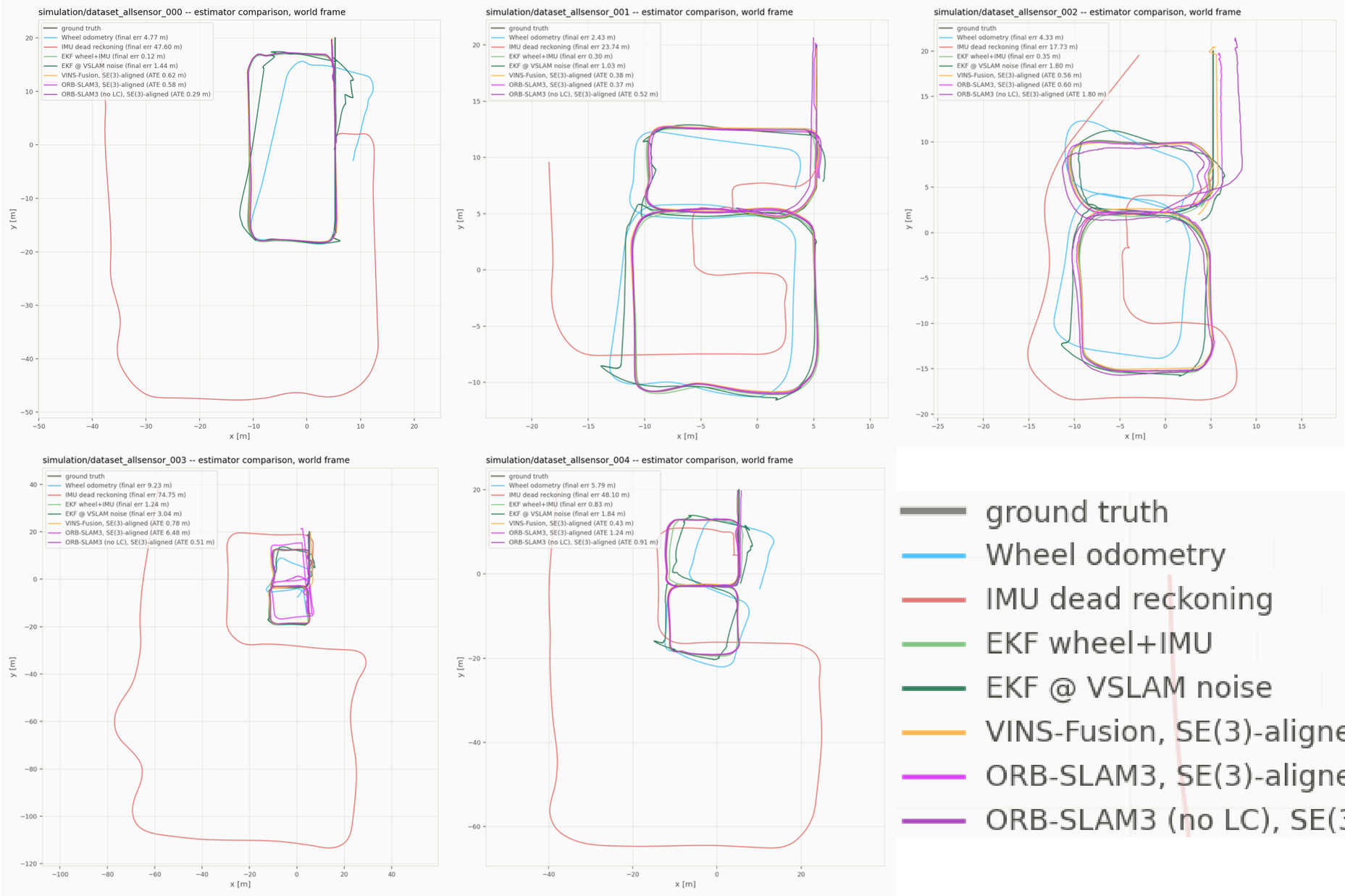


5. Experimental Design

Experiment 1 : Non-Visual Baseline and Estimator Comparison

- Relative pose error (RPE) at delta time = 1 second

Estimator	Trans. RPE (m)	Rot. RPE (°)
Wheel odometry	**0.074**	2.777
IMU dead reckoning	**3.558**	0.042
EKF	0.276	1.371
VINS-Fusion	0.103	**0.041**
ORB-SLAM3	0.121	0.117


simulation/dataset_allsensor_000 -- estimator comparison, world frame
ground truth
Wheel odometry (final err 4.77 m)
IMU dead reckoning (final err 47.60 m)
EKF wheel+IMU (final err 0.12 m)
EKF @ VSLAM noise (final err 1.44 m)
VINS-Fusion, SE(3)-aligned (ATE 0.62 m)
ORB-SLAM3, SE(3)-aligned (ATE 0.58 m)
ORB-SLAM3 (no LC), SE(3)-aligned (ATE 0.29 m)
simulation/dataset_allsensor_001 -- estimator comparison, world frame
ground truth
Wheel odometry (final err 2.43 m)
IMU dead reckoning (final err 23.74 m)
EKF wheel+IMU (final err 0.30 m)
EKF @ VSLAM noise (final err 1.03 m)
VINS-Fusion, SE(3)-aligned (ATE 0.38 m)
ORB-SLAM3, SE(3)-aligned (ATE 0.37 m)
ORB-SLAM3 (no LC), SE(3)-aligned (ATE 0.52 m)
simulation/dataset_allsensor_002 -- estimator comparison, world frame
ground truth
Wheel odometry (final err 4.33 m)
IMU dead reckoning (final err 17.73 m)
EKF wheel+IMU (final err 0.35 m)
EKF @ VSLAM noise (final err 1.80 m)
VINS-Fusion, SE(3)-aligned (ATE 0.56 m)
ORB-SLAM3, SE(3)-aligned (ATE 0.60 m)
ORB-SLAM3 (no LC), SE(3)-aligned (ATE 1.80 m)
simulation/dataset_allsensor_003 -- estimator comparison, world frame
ground truth
Wheel odometry (final err 9.23 m)
IMU dead reckoning (final err 74.75 m)
EKF wheel+IMU (final err 1.24 m)
EKF @ VSLAM noise (final err 3.04 m)
VINS-Fusion, SE(3)-aligned (ATE 0.78 m)
ORB-SLAM3, SE(3)-aligned (ATE 6.48 m)
ORB-SLAM3 (no LC), SE(3)-aligned (ATE 0.51 m)
simulation/dataset_allsensor_004 -- estimator comparison, world frame
ground truth
Wheel odometry (final err 5.79 m)
IMU dead reckoning (final err 48.10 m)
EKF wheel+IMU (final err 0.83 m)
EKF @ VSLAM noise (final err 1.84 m)
VINS-Fusion, SE(3)-aligned (ATE 0.43 m)
ORB-SLAM3, SE(3)-aligned (ATE 1.24 m)
ORB-SLAM3 (no LC), SE(3)-aligned (ATE 0.91 m)
x [m]
y [m]
ground truth
Wheel odometry
IMU dead reckoning
EKF wheel+IMU
EKF @ VSLAM noise
VINS-Fusion, SE(3)-aligned
ORB-SLAM3, SE(3)-aligned
ORB-SLAM3 (no LC), SE(3)-aligned

5. Experimental Design

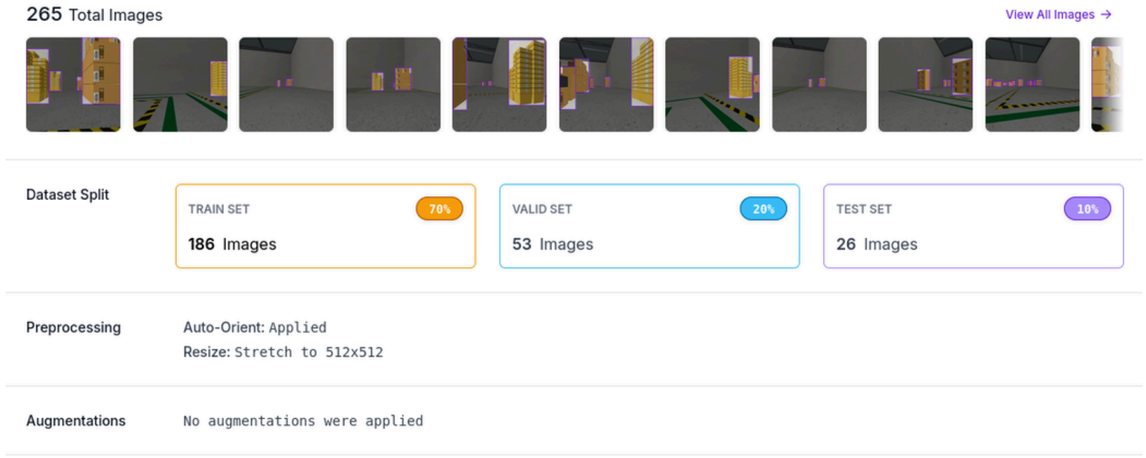
Experiment 1 : Non-Visual Baseline and Estimator Comparison

- **Summary**
  - IMU-only 12.6–45.6 m, wheels 1.0–3.2 m.
  - EKF < 0.5 m.
  - VINS/ORB < 0.2 m
- **Future Work**
  - Feed VIO into the EKF as a third measurement, so the filter takes distance from wheels, heading from the gyro, and drift correction from vision.
  - Re-run in the realistic environment (wheel slip, encoder quantisation), both idealisations currently favour dead reckoning.

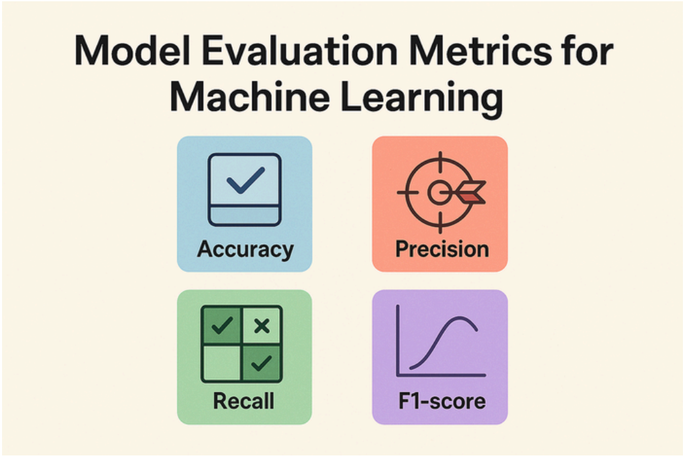
5. Experimental Design

Experiment 2 : Offline YOLO Detector Performance

- Objective : Measure standalone detector performance independent of SLAM so future filtering results are attributed to mask logic rather than detector capabilities.
- Process :



1.Annotation the images



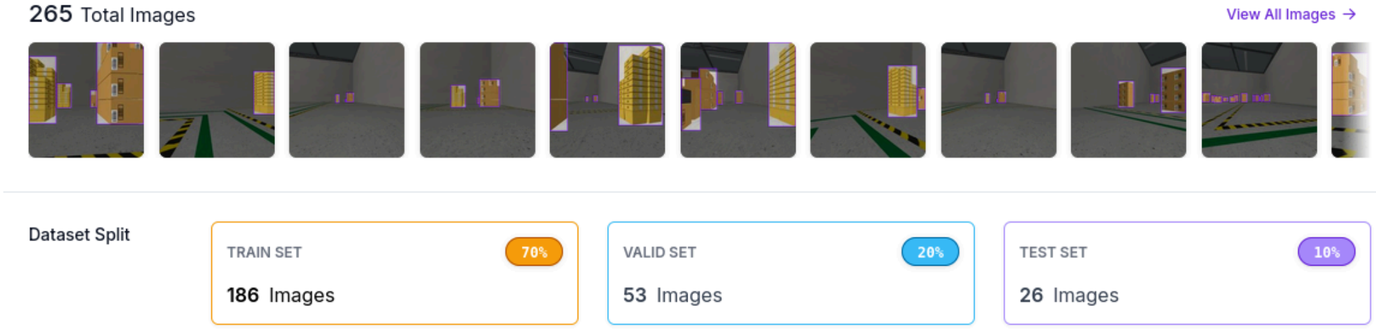
2. Model Training and Evaluation

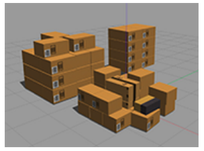
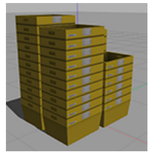
- Evaluation Metrics :
  - Accuracy
  - Precision, Recall
  - Image-Area Coverage

5. Experimental Design

Experiment 2 : Offline YOLO Detector Performance

- Dataset Information

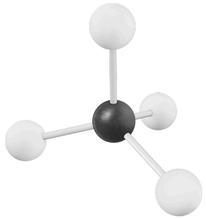


	Class name	Count
	bin	34
	box	1030
	bucket	243

5. Experimental Design

Experiment 2 : Offline YOLO Detector Performance

- Object Ground Truth in Simulation



Black dot pose : ros2 topic → /world/default/pose/info
White dot : Mesh files

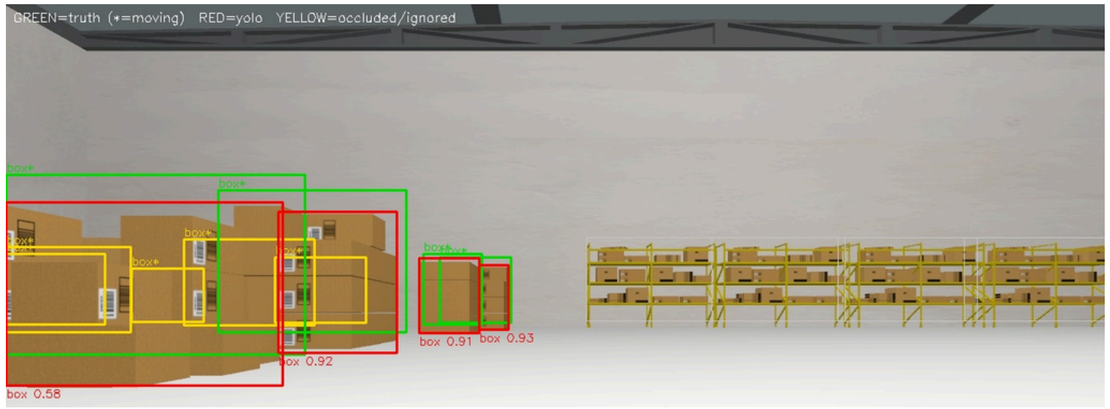
$$v_w = R_{wo}V + t_{wo}$$

Sets of Vertices pose (World frame)

$$T_{wc} = T_{wb}T_{bc}$$

Sets of Vertices pose (Camera frame)

$$R_{cw} = R_{wc}^T$$
$$t_{cw} = -R_{cw}t_{wc}$$
$$v_c = R_{cw}v_w + t_{cw}; v_c = \{x_c, y_c, z_c\}$$



Projection :

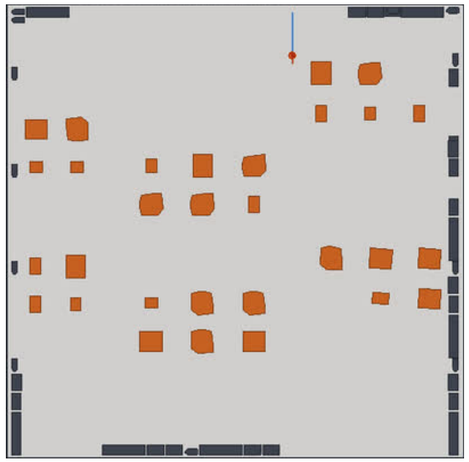
$$u = f_x\frac{x_c}{z_c} + c_x \quad v = f_y\frac{y_c}{z_c} + c_y$$

5. Experimental Design

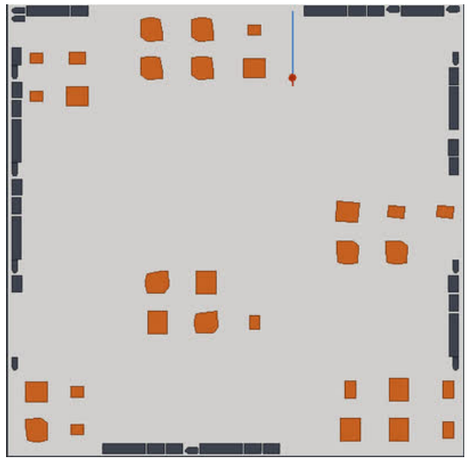
Experiment 2 : Offline YOLO Detector Performance

- Dataset information

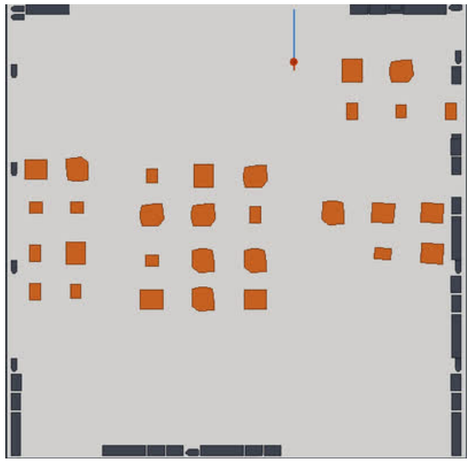
Dataset	Family	Size (MB)	Duration. (s)	Messages	/clock	Truth	Prop.
dataset_dynamic_nofloortexture_00_000	dynamic_00	653	100.3	111 988	✓	✓	–
dataset_dynamic_nofloortexture_00_001	dynamic_00	778	167.7	114 169	✓	✓	–
dataset_dynamic_nofloortexture_01_000	dynamic_01	713	165.4	126 790	✓	✓	–
dataset_dynamic_nofloortexture_02_000	dynamic_02	758	172.1	134 995	✓	✓	–



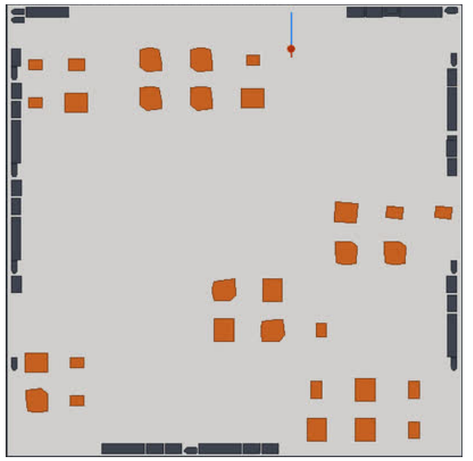
dynamic_00_000



dynamic_01_000



dynamic_01_000



dynamic_02_000

5. Experimental Design

Experiment 2 : Offline YOLO Detector Performance

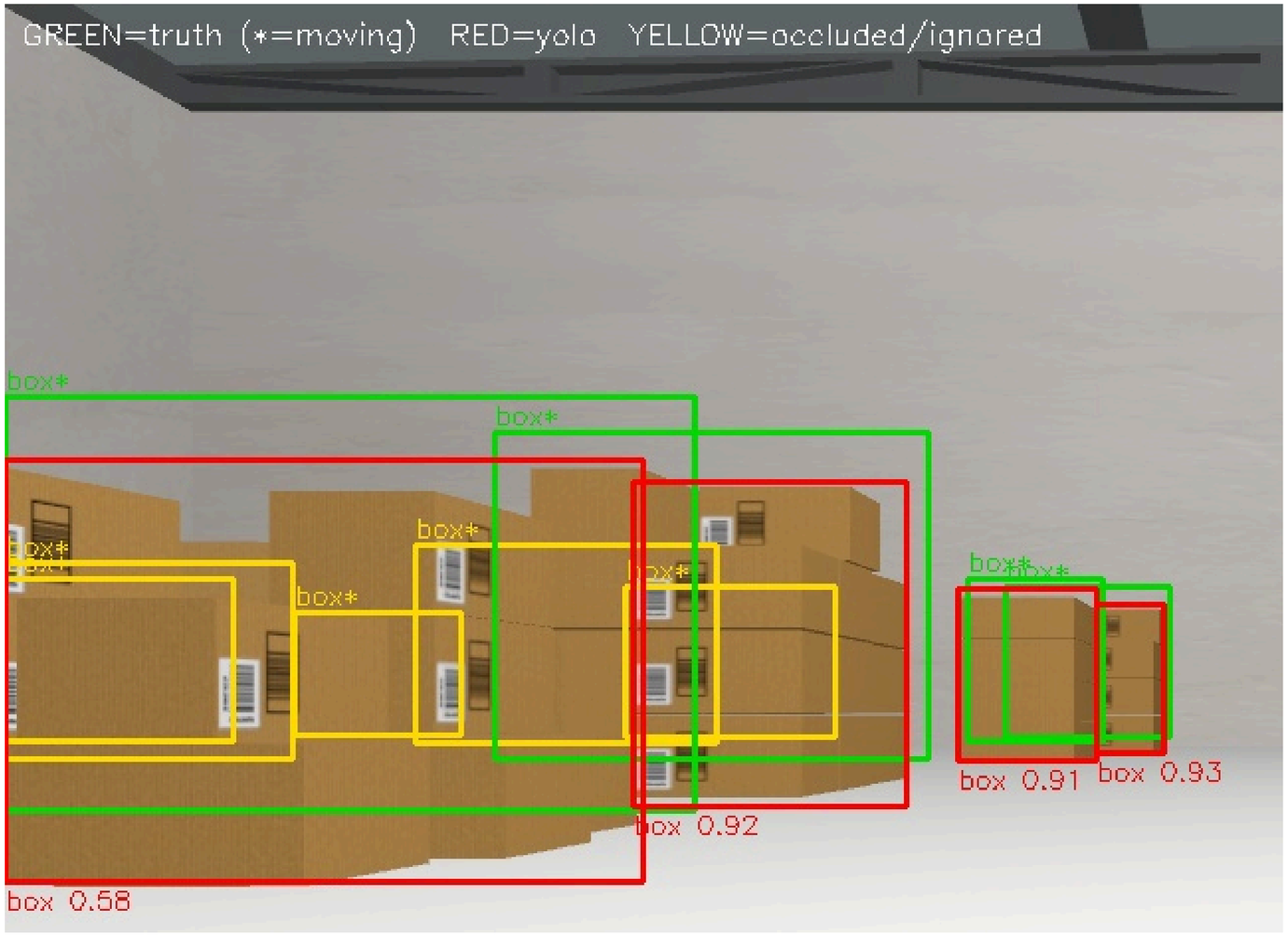
- **Detection Result**

Dataset	Frames	TP	FP	FN	Precision	Recall	F1	Moving recall
Dynamic 00/000	2,347	9,987	4,826	5,900	0.674	0.629	0.651	0.617
Dynamic 00/001	2,236	10,158	4,265	6,868	0.704	0.597	0.646	0.599
Dynamic 01/000	2,564	8,142	4,226	6,064	0.658	0.573	0.613	0.568
Dynamic 02/000	2,713	8,661	3,743	6,502	0.698	0.571	0.628	0.581
Pooled	**9,860**	**36,948**	**17,060**	**25,334**	**0.684**	**0.593**	**0.635**	**0.591**

5. Experimental Design

Experiment 2 : Offline YOLO Detector Performance

- **Detection Result**



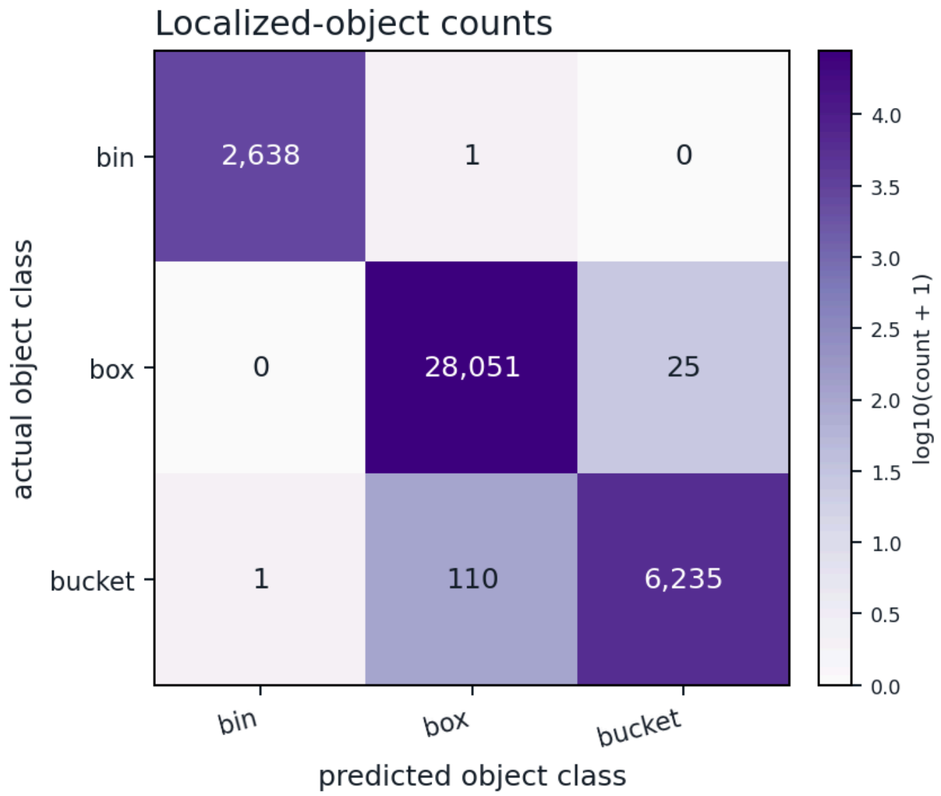
5. Experimental Design

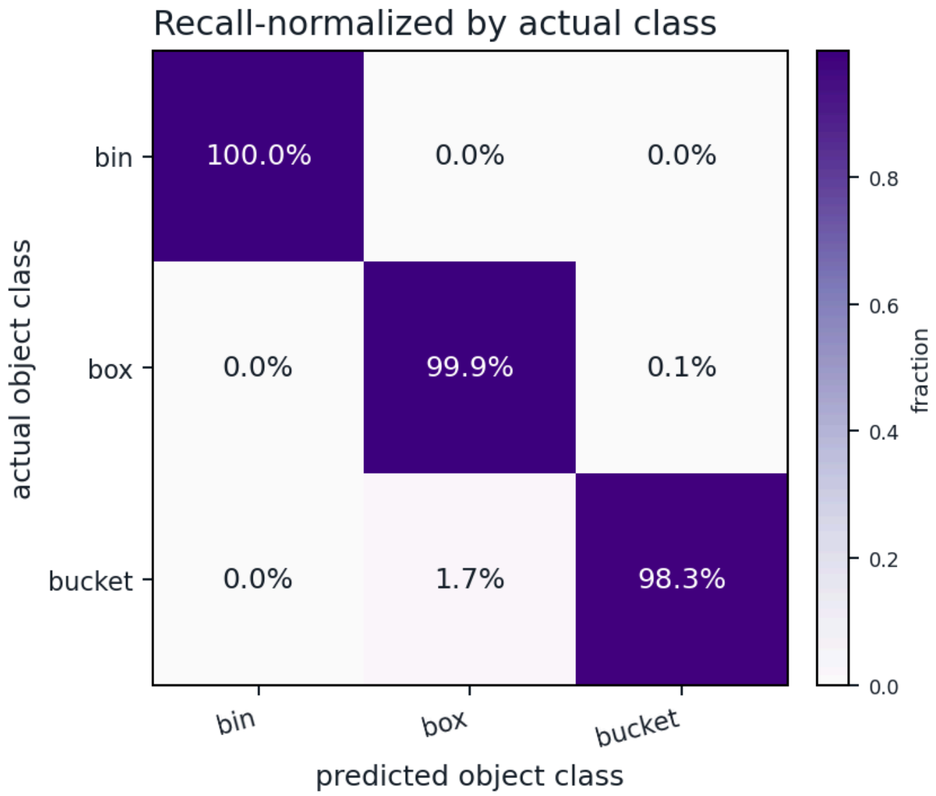
Experiment 2 : Offline YOLO Detector Performance

- **Object Classification Result**

Class	Associated objects	Precision	Recall	F1
Bin	2,639	1	1	1
Box	28,076	0.996	0.999	0.998
Bucket	6,346	0.996	0.983	0.989
Macro average	37,061	0.997	0.994	0.995

YOLO object-class assignment, conditional on IoU-matched localization





5. Experimental Design

Experiment 2 : Offline YOLO Detector Performance

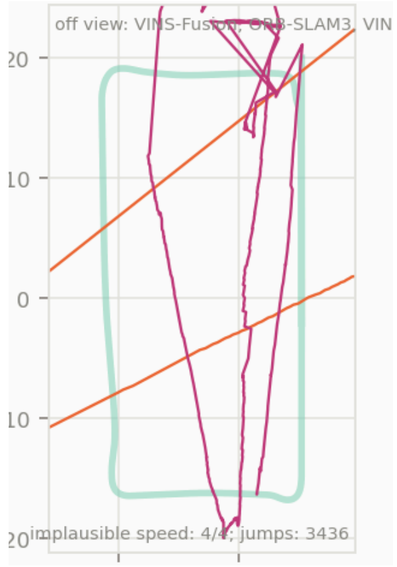
- **Summary**
  - The detector is reliable at naming and unreliable at finding: 0.68 / 0.59 / 0.995 (P / moving-R / class-F1).
  - The 41% of moving objects it misses is the limit any downstream filter inherits.
- **Future Plan**
  - Validate the projected object ground truth against manual annotation on a frame subset.
  - Add object velocity from consecutive detections so the mask targets moving objects rather than object classes.
  - Re-run on the realistic warehouse environment once built.

5. Experimental Design

Experiment 3: Visual-Inertial Odometry With Dynamic Environment Filtering Evaluation

- Objective : Determine if masking dynamic-object features improves localization and measure the associated computational costs.
- Process :

1. Apply VSLAM approach from methodology (3 times per set)



2. Result Evaluation

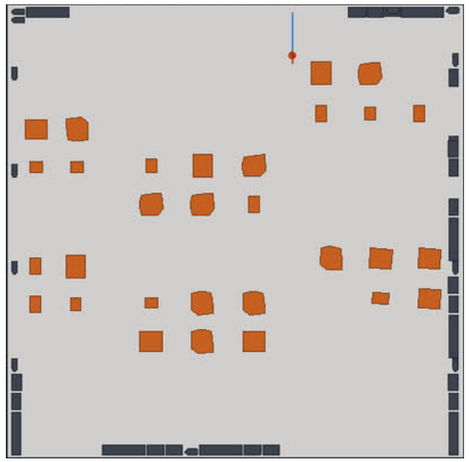
- Evaluation Metrics :
  - Trajectory Accuracy
  - Algorithm Robustness
  - Algorithm Latency.

5. Experimental Design

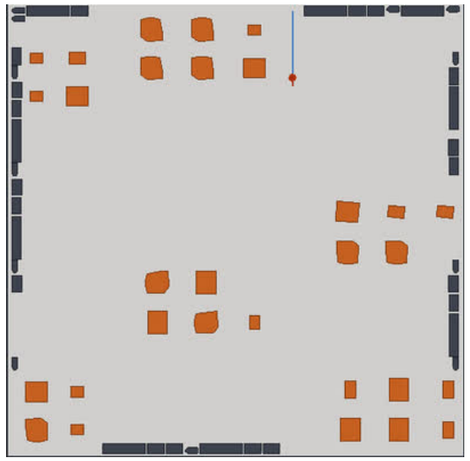
Experiment 3: Visual-Inertial Odometry With Dynamic Environment Filtering Evaluation

- Dataset information

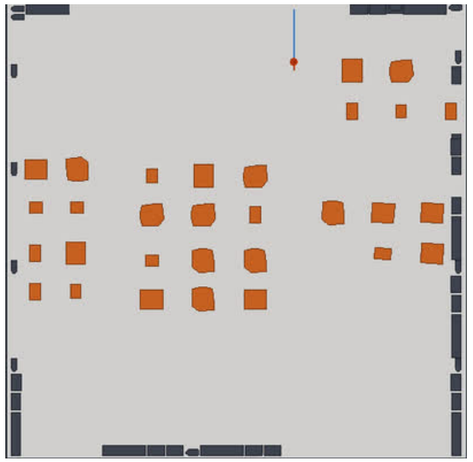
Dataset	Family	Size (MB)	Duration. (s)	Messages	/clock	Truth	Prop.
dataset_dynamic_nofloortexture_00_000	dynamic_00	653	100.3	111 988	✓	✓	–
dataset_dynamic_nofloortexture_00_001	dynamic_00	778	167.7	114 169	✓	✓	–
dataset_dynamic_nofloortexture_01_000	dynamic_01	713	165.4	126 790	✓	✓	–
dataset_dynamic_nofloortexture_02_000	dynamic_02	758	172.1	134 995	✓	✓	–



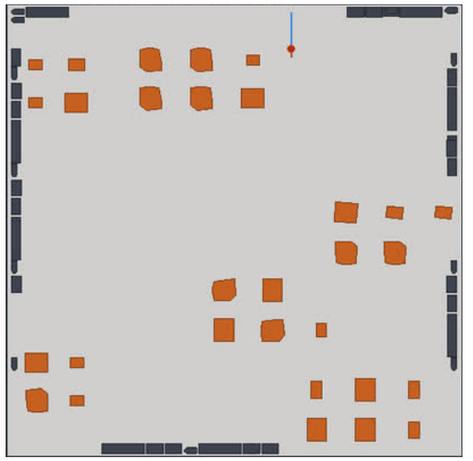
dynamic_00_000



dynamic_01_000



dynamic_01_000



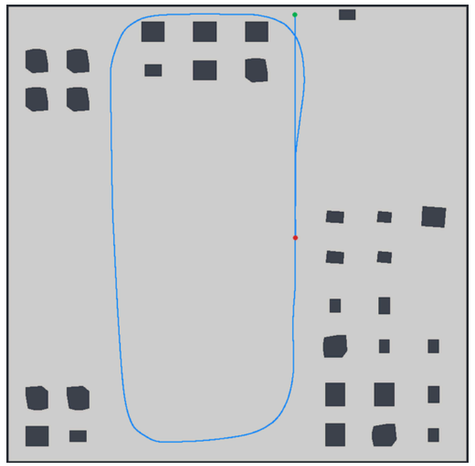
dynamic_02_000

5. Experimental Design

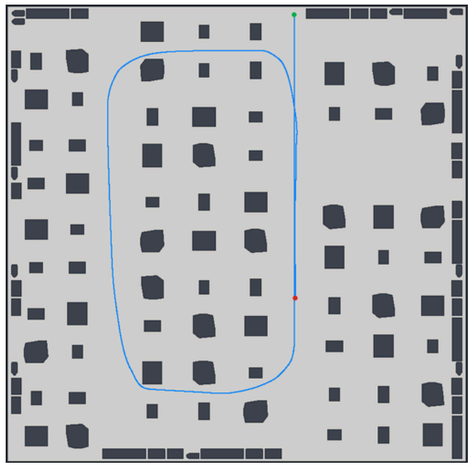
Experiment 3: Visual-Inertial Odometry With Dynamic Environment Filtering Evaluation

- Dataset information

Dataset	Family	Size (MB)	Duration. (s)	Messages	/clock	Truth	Prop.
dataset_static_nofloortexture_000	static	592	101.5	118 597	✓	✓	–
dataset_static_nofloortexture_001	static	617	92	113 332	✓	✓	–
dataset_static_nofloortexture_002	static	688	103.3	122 386	✓	✓	–



static_000



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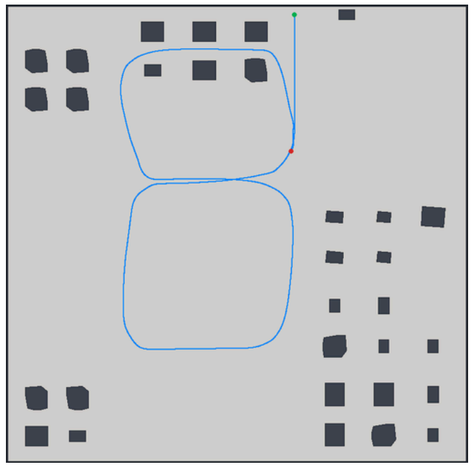
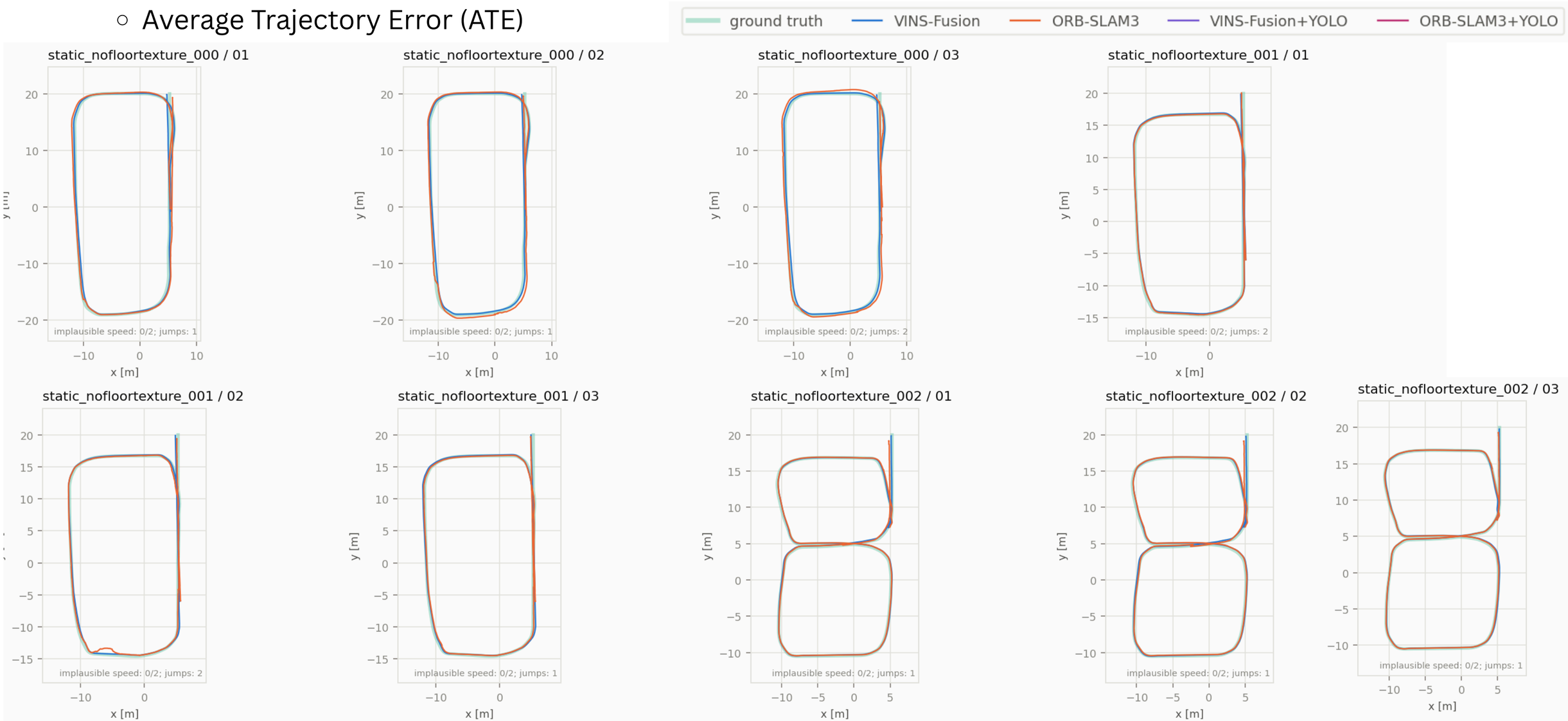


static_002

5. Experimental Design

Experiment 3: Visual-Inertial Odometry With Dynamic Environment Filtering Evaluation

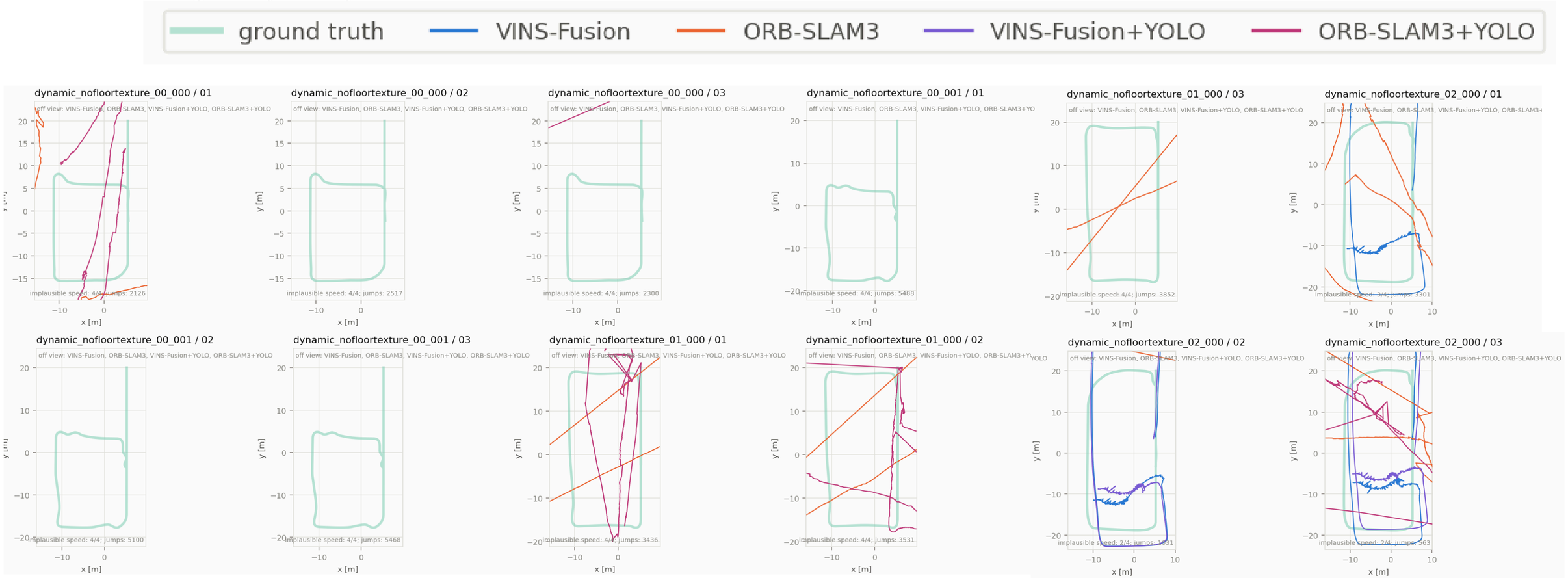
- Average Trajectory Error (ATE)



5. Experimental Design

Experiment 3: Visual-Inertial Odometry With Dynamic Environment Filtering Evaluation

- Average Trajectory Error (ATE)



5. Experimental Design

Experiment 3: Visual-Inertial Odometry With Dynamic Environment Filtering Evaluation

- Average Trajectory Error (ATE)

Scene	Requested mode	ATE RMSE (m)	Rotation RMSE (deg)	Clean ATE (m)
Static	VINS-Fusion	0.37	1.56	0.37
Static	ORB-SLAM3	0.454	17.92	0.341
Dynamic	VINS-Fusion	596.367	38.51	6.771
Dynamic	ORB-SLAM3	45.206	176.53	1.314
Dynamic	ORB-SLAM3+YOLO	13.839	76.72	1.31

- Relative pose error (RPE) at delta time = 1 second

Scene	Requested mode	Jumps	Translational (m) all pairs	Rotational (deg) all pairs
Static	VINS-Fusion	0	0.063	0.57
Static	ORB-SLAM3	1	0.31	10.56
Dynamic	VINS-Fusion	1116	44.652	2.05
Dynamic	ORB-SLAM3	484	69.477	21.6
Dynamic	ORB-SLAM3+YOLO	208	38.174	15.15

5. Experimental Design

Experiment 3: Visual-Inertial Odometry With Dynamic Environment Filtering Evaluation

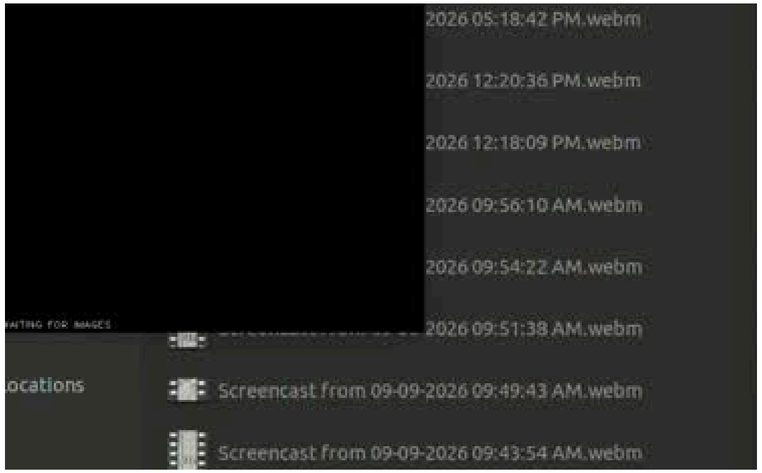
- Dynamic-run capacity and resource measurements

Requested mode	Throughput (Hz)	Retained input (%)	CPU (% core)	Peak RSS (MiB)
VINS-Fusion	27.14 ± 5.81	100.00 ± 0.00	443.66 ± 58.96	629.51 ± 282.62
ORB-SLAM3	23.47 ± 4.77	88.55 ± 13.42	385.22 ± 27.71	1226.06 ± 81.83
ORB-SLAM3+YOLO	14.47 ± 5.42	66.96 ± 15.26	410.86 ± 42.83	1054.64 ± 91.11

5. Experimental Design

Experiment 3: Visual-Inertial Odometry With Dynamic Environment Filtering Evaluation

- Analysis : Filter Capability



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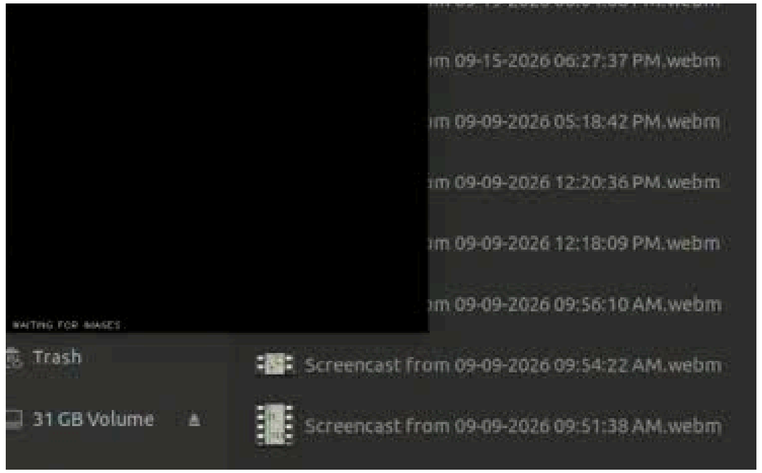

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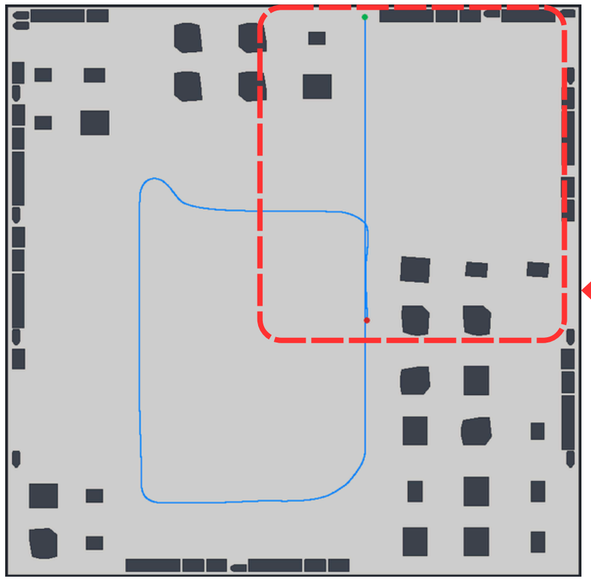
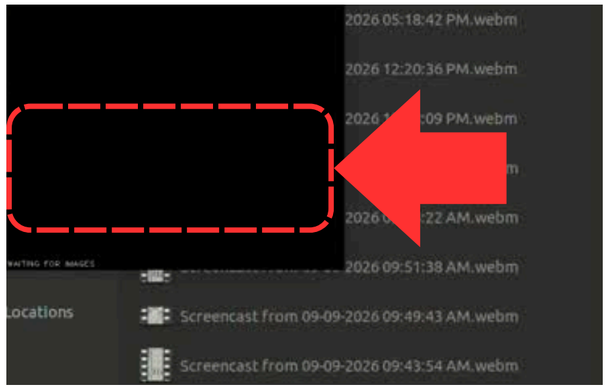


Ground Truth Trajectory

5. Experimental Design

Experiment 3: Visual-Inertial Odometry With Dynamic Environment Filtering Evaluation

- Analysis : Straight Path



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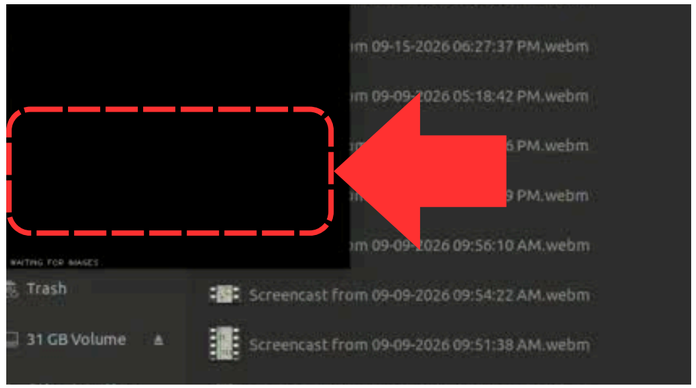

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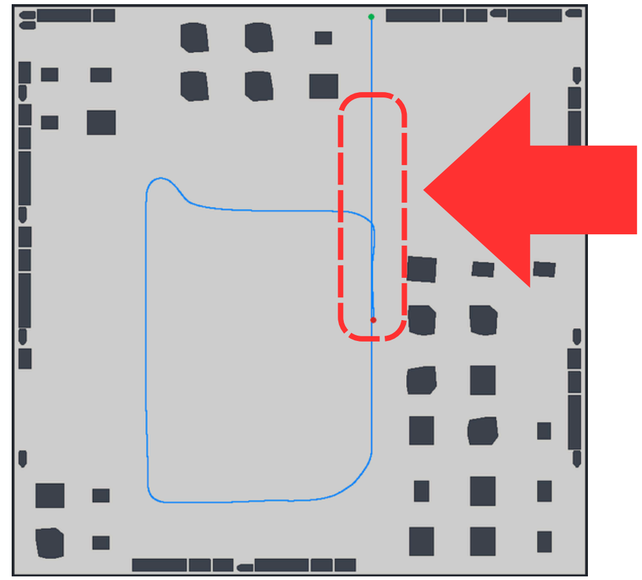
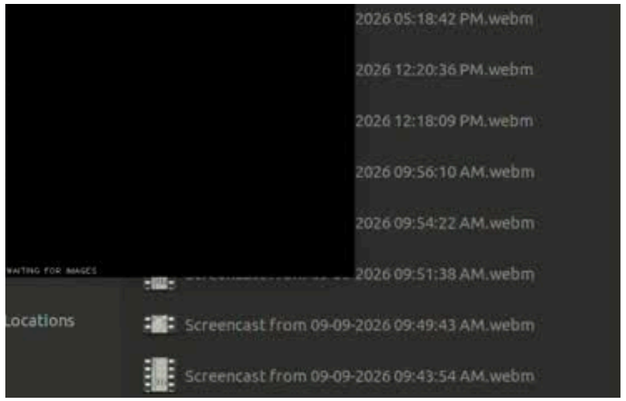


Ground Truth Trajectory

5. Experimental Design

Experiment 3: Visual-Inertial Odometry With Dynamic Environment Filtering Evaluation

- Analysis : Heading-change Path



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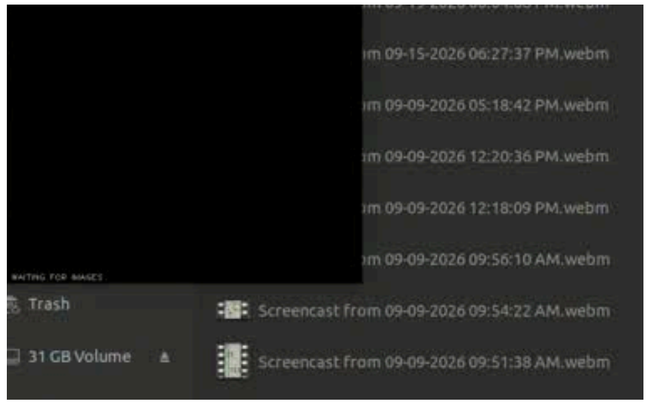

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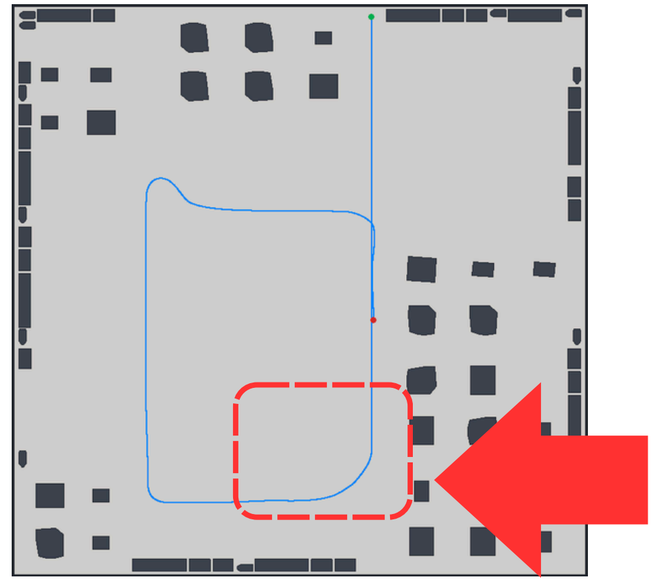


Ground Truth Trajectory

5. Experimental Design

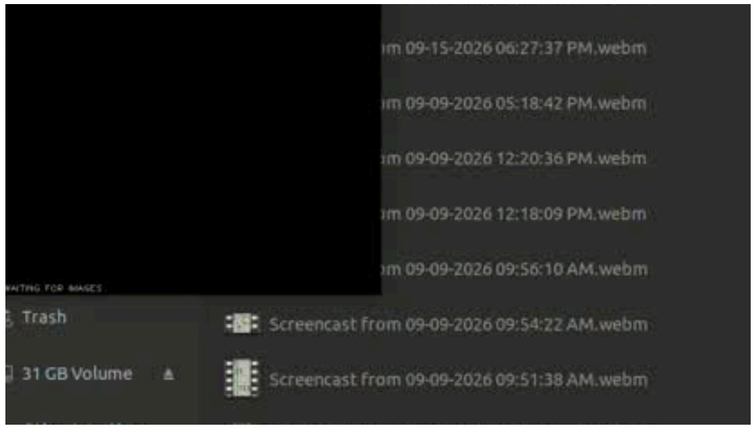
Experiment 3 : Visual-Inertial Odometry With Dynamic Environment Filtering Evaluation

- **Summary**
  - The core hypothesis has no valid test yet: one pipeline's filter never fired, the other's comparison is confounded by frame loss, and the environment is not physical.
  - This is untested, not tested and failed.
- **Future Plan**
  - Fix the detection→VINS integration so masked tracks are actually culled; verify by the removed-track counter before any re-run.
  - Reduce ORB-SLAM3+YOLO26 latency so filtered and unfiltered runs see the same frames.
  - Rebuild the dynamic worlds with physically simulated props.

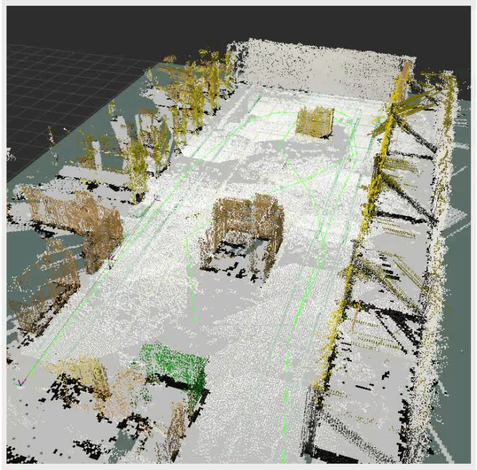
5. Experimental Design

Experiment 4: Dense Map Reconstruction and Map-Quality Evaluation

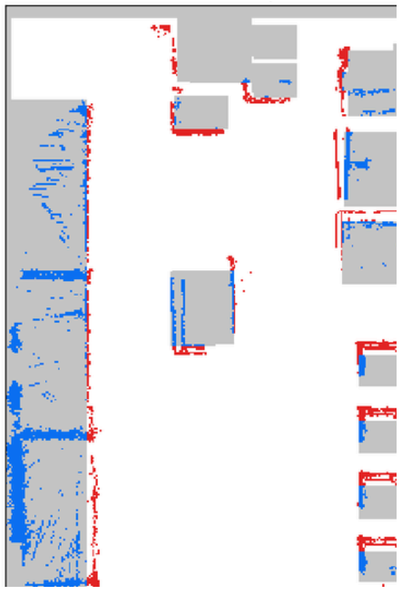
- Objective : Verify if the reconstructed map accurately reflects the physical environment and is usable by a route planner.
- Process :



1.Map Collection



2.Apply map



3.Map Evaluation

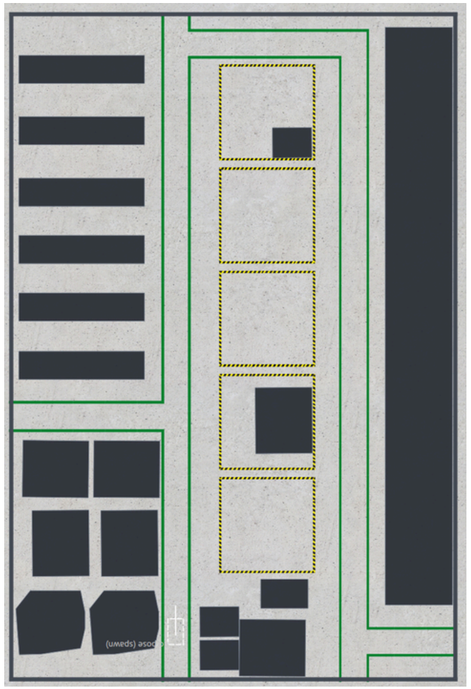
- Evaluation Metrics :
  - Area Coverage Precision and Recall
  - False-Obstacle Percentage.

5. Experimental Design

Experiment 2 : Offline YOLO Detector Performance

- Dataset information

Dataset	Family	Size (MB)	Duration. (s)	Messages	/clock	Truth	Prop.
dataset_map_tiny	mapping, tiny world	726	63.3	82 555	✓	✓	–

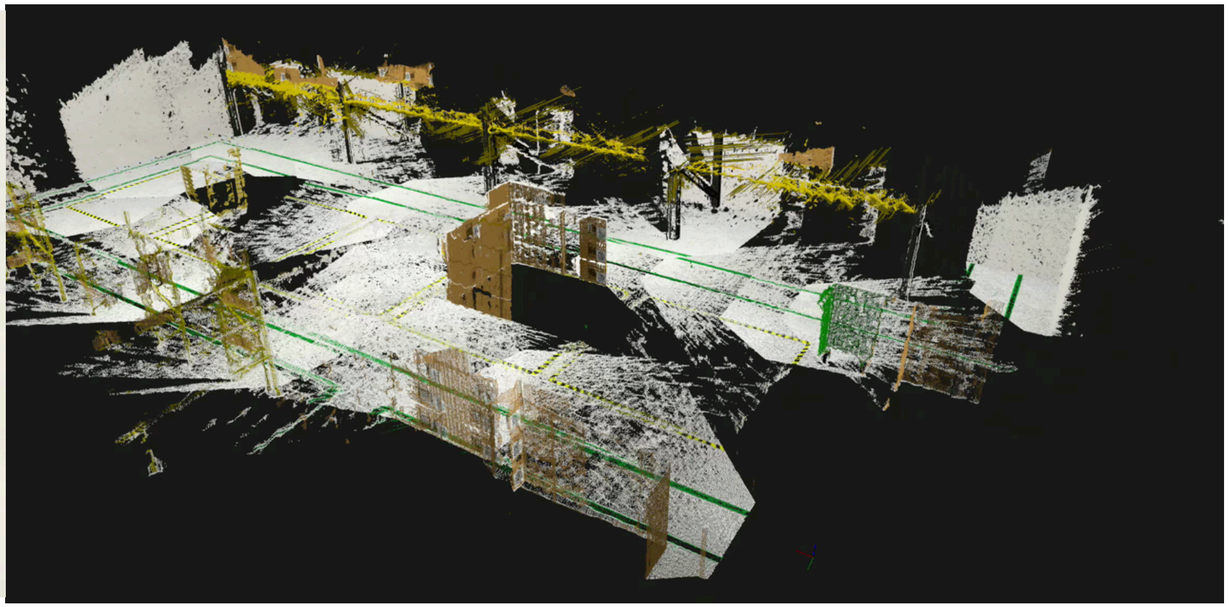


dataset_map_tiny

5. Experimental Design

Experiment 4: Dense Map Reconstruction and Map-Quality Evaluation

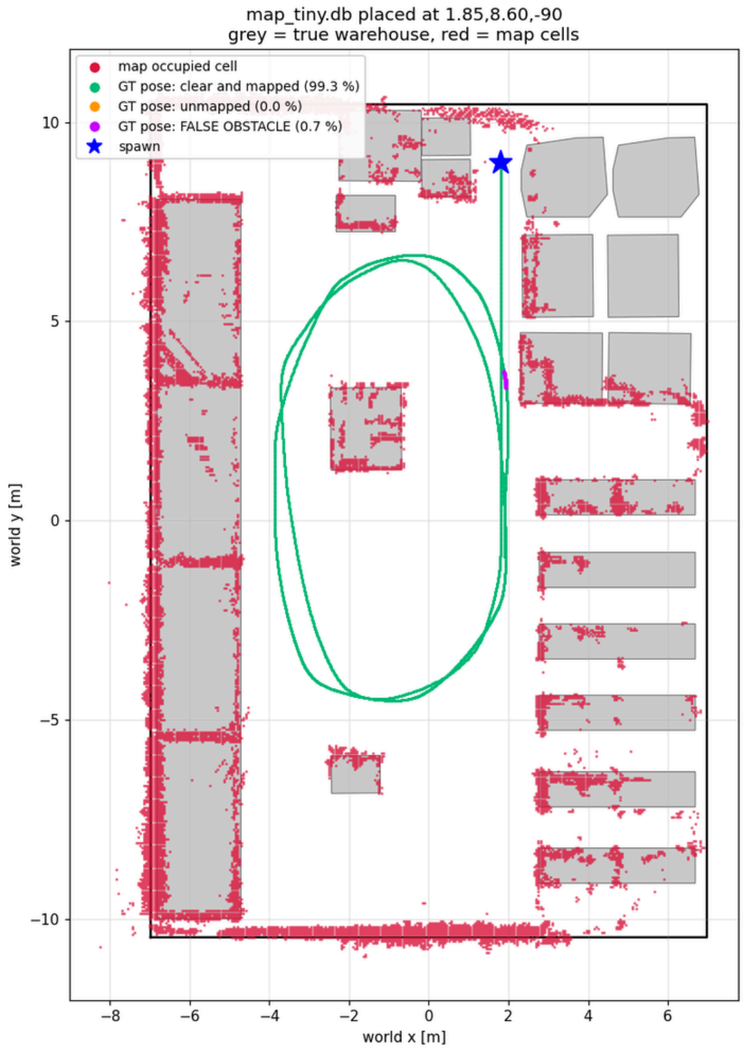
- 3D Map Construction

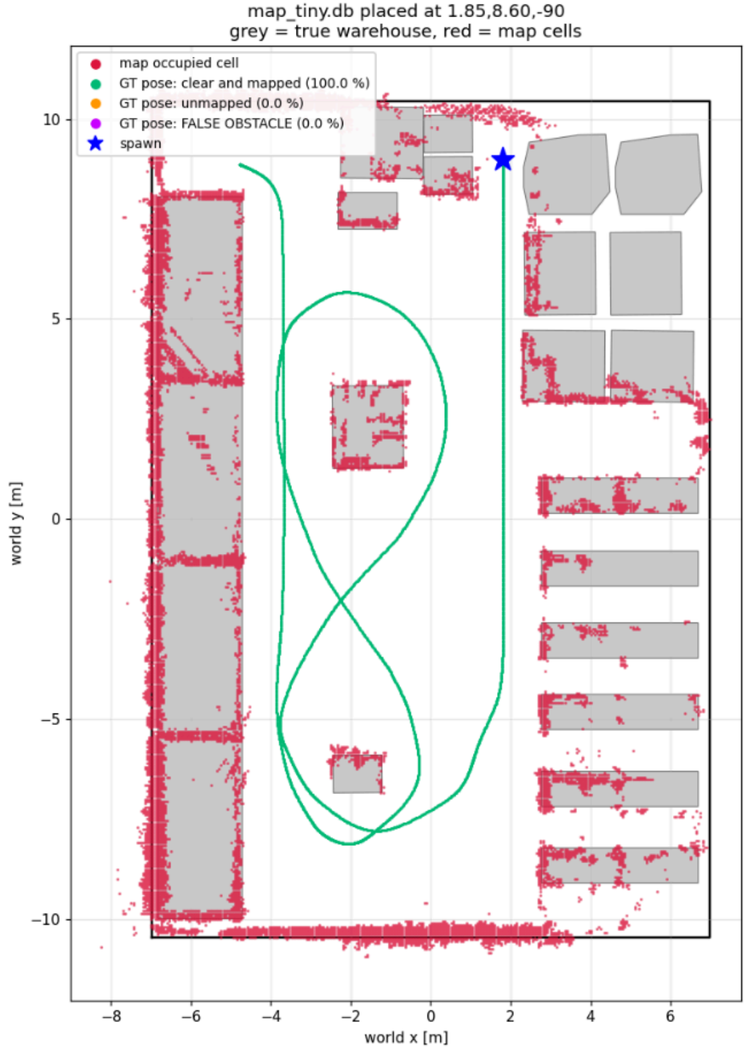


5. Experimental Design

Experiment 4: Dense Map Reconstruction and Map-Quality Evaluation

- Result : tiny map





Precision	Recall
96.6	67.7

5. Experimental Design

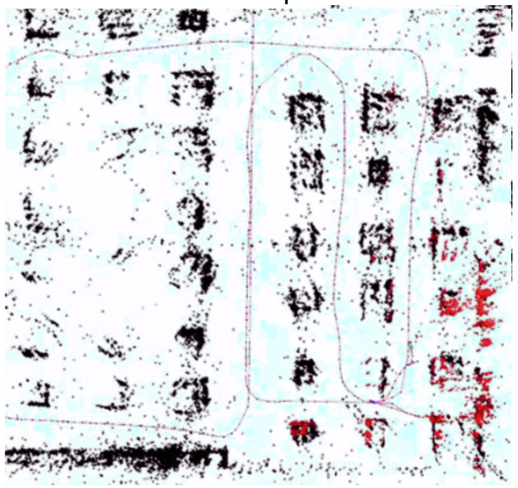
Experiment 4: Dense Map Reconstruction and Map-Quality Evaluation

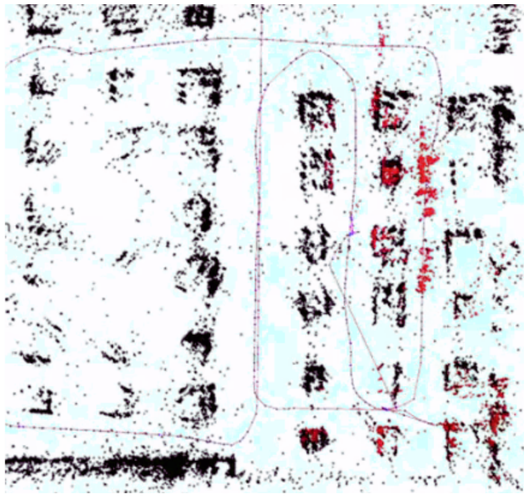
- Result : Large map

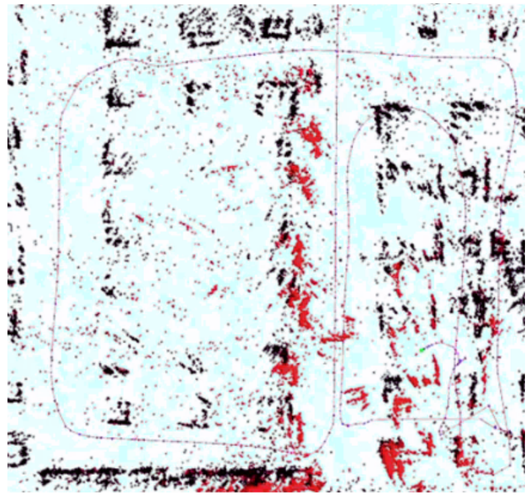
timestep t

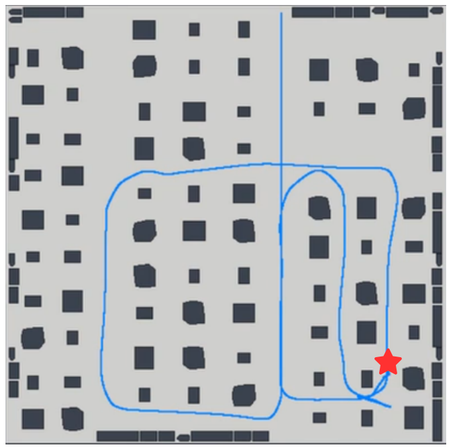
timestep t+1

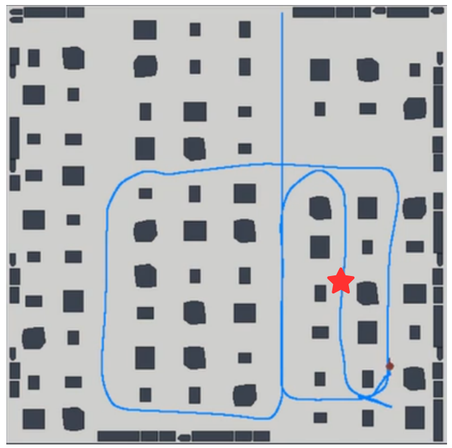
timestep t+2

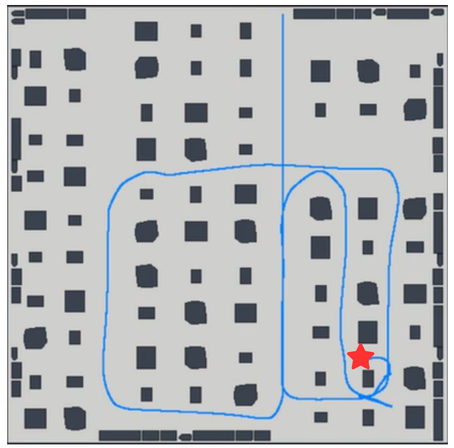












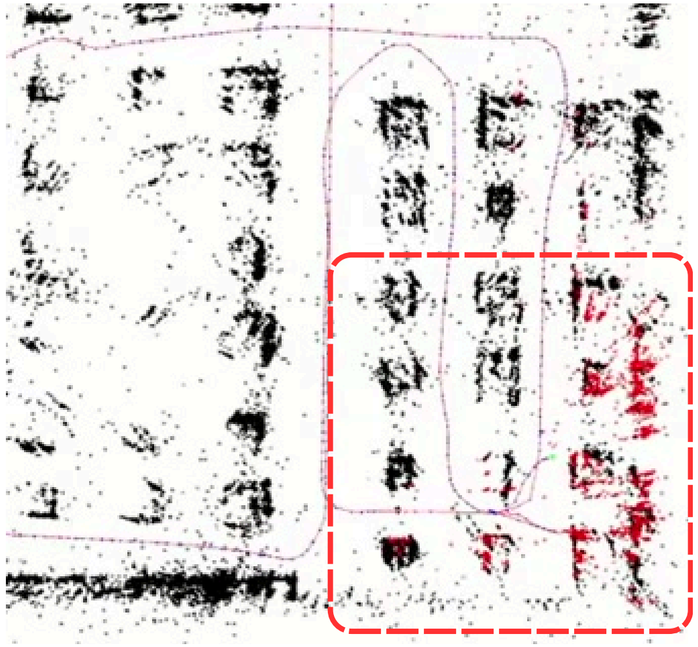
: ORB's pose

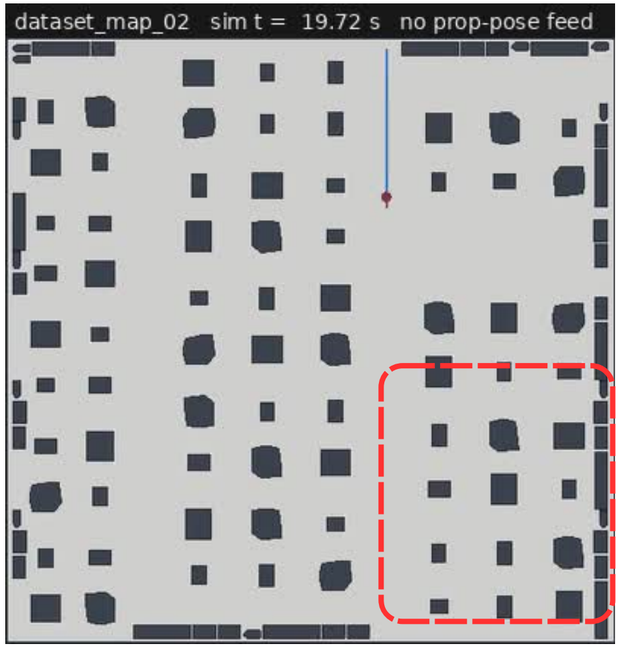
: Ground truth trajectory

5. Experimental Design

Experiment 4: Dense Map Reconstruction and Map-Quality Evaluation

- Result : Large map





5. Experimental Design

Experiment 4 : Dense Map Reconstruction and Map-Quality Evaluation

- **Summary**
  - The mapping pipeline is geometrically sound where tracking is sound (tiny map, 96.6% / 89.5%).
  - Whole-building recall of 67% is route coverage, side aisles were never entered, not mapping failure.
  - It fails on the large map through repetitive structure, not through the mapping back end.
- **Future Plan**
  - Study methods robust to repeated features (place-recognition gating, geometric verification of loop candidates).
  - Enable ORB-SLAM3 Atlas so a tracking loss opens a new sub-map instead of corrupting the existing one; merge sub-maps afterwards.
  - Record an independent route on both maps and score drivability against it.

6. Future Plan

Done

Current Progress

แผนดำเนินงาน	มิ.ย.	ก.ค.	ส.ค.	ก.ย.	ต.ค.	พ.ย.	ธ.ค.
1. ศึกษาแนวคิดและทฤษฎีที่เกี่ยวข้องกับระบบการนำทางด้วยภาพ (Vision-Based Navigation)	✓	✓					
2. สืบค้นและวิเคราะห์ข้อมูลเกี่ยวกับอัลกอริทึม Visual SLAM, ระบบปรับปรุงคุณภาพภาพในสภาวะแสงน้อย (Image Enhancement: EnlightenGAN), ระบบตรวจจับวัตถุ (Object Detection: YOLO26)	✓	✓					
3. ออกแบบโครงสร้างและไปป์ไลน์การทำงานร่วมกันระหว่างระบบฮาร์ดแวร์และซอฟต์แวร์ (Software Pipeline)		✓	✓				
4. กำหนดคุณสมบัติและการเชื่อมต่อของเซนเซอร์กล้อง AC-IMX390-H190 ร่วมกับหน่วยประมวลผลระดับเอดจ์ (Edge Computing) รุ่น NRU-51V บนตัวหุ่นยนต์ Mobile Robot		✓	✓				
6. บูรณาการระบบย่อยทั้งหมดเข้าด้วยกันภายใต้กรอบการทำงานระบบนำทางของหุ่นยนต์ AMR			✓	✓			
7. จัดเตรียมพื้นที่คลังสินค้าจำลอง (Zone A และ Zone B) ที่มีความลาดชัน 5 องศา และจัดวางพัสดุจำลองเพื่อสร้างสภาพแวดล้อมแบบพลวัต (Dynamic Environment)				✓	✓		
8. ดำเนินการทดสอบระบบแบบบูรณาการ (Integrated System Experiment / Live Test) โดยให้หุ่นยนต์เคลื่อนที่ด้วยความเร็ว 1.5 เมตรต่อวินาที ภายใต้เงื่อนไขการเปลี่ยนแปลงของแสงสว่างตามช่วงเวลาที่กำหนด (Simulation and real world)				✓	✓		
9. รวบรวมข้อมูลจากการทดสอบมาประเมินผลประสิทธิภาพของระบบตามตัวชี้วัดที่กำหนด เช่น ค่าความคลาดเคลื่อนของเส้นทาง (ATE, RPE) ความหน่วงของระบบ (Latency) และความแม่นยำในการตรวจจับวัตถุ (IoU, mAP)					✓	✓	
10. สรุปผลการดำเนินงาน อภิปรายผล และจัดทำเล่มรายงานโครงงานวิศวกรรมฉบับสมบูรณ์						✓	✓

Thank you

Contact Us

Institute of Field Robotics
King Mongkut's University of Technology Thonburi

A Cradle of Future Leaders in Robotics